LOYOLA ACADEMY

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Practical Record <u>CERTIFICATE</u>

This is to certify that this is a bonafide record work done in **DBMS PROGRAMMING Practical**

During 3rd year 1st semester of the academic year 2025-2026

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STRUCTUREDQUERYLANGUAGE

1) **WRITE A QUERY TO CREATE A TABLE WITH THE FOLLOWINGPARAMETERSSNO,SNAME,DOB,SUB1,SUB2,AND SUB3 USING A APPROPRIATE DATA TYPE.**

SQL>CREATETABLESTUD1(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>DESC

STUD1

```
Table created.

SQL> desc stud1
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                                NUMBER(11)
 SNAME                                     NOT NULL VARCHAR2(13)
 DOB                                                DATE
 SUB1                                               CHAR(8)
 SUB2                                               CHAR(8)
 SUB3                                               CHAR(8)
```

2) WRITEAQUERYTOINSERTDATAINTOTABLEANDDISPLAYTHETABLE.

SQL>INSERTINTOSTUD1VALUES(&SNO,’&SNAME’,’&DOB’,’&SUB1’,’2&SUB2’,’&SUB3’);

```
SQL> insert into stud1 values(&sno,’&sname’,’&dob’,’&sub1’,’2 &sub2’,’&sub3’);
Enter value for sno: 1
Enter value for sname: sailaja
Enter value for dob: 3-oct-1994
Enter value for sub1: Eng
Enter value for sub2: Litera
Enter value for sub3: Phylo
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(1,'sailaja','3-oct-1994','Eng','2 Litera','Phylo')

1 row created.

SQL> /
Enter value for sno: 2
Enter value for sname: shanmukh
Enter value for dob: 11-mar-1999
Enter value for sub1: phy
Enter value for sub2: ldc
Enter value for sub3: m1
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(2,'shanmukh','11-mar-1999','phy','2 ldc','m1')

1 row created.
SQL> /
Enter value for sno: 3
Enter value for sname: hardik jain
Enter value for dob: 02-oct-2005
Enter value for sub1: engg
Enter value for sub2: m2
Enter value for sub3: pis
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(3,'hardik jain','02-oct-2005','engg','2 m2','pis')

1 row created.
```

SQL>SELECT*FROMSTUD1;

```
SQL> select * from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

3) **WRITEAQUERYTODISPLAYTHEDATAINTABLEASI.SELECTROWSAND COLUMNS II. SELECTED COLUMNS AND ALL ROWS ALL ROWS III. ALL ROWS AND ALL COLUMNS IV. SELECTED ROWS AND ALL COLUMNS**

I. SQL>SELECT*FROMSTUD1WHERESNO>2;

```
SQL> select * from stud1 where sno>2;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

II. SQL>SELECTSNO,SNAMEFROMSTUD1;

```
SQL> select sno,sname from stud1;

       SNO SNAME
---------- -------------
         1 sailaja
         2 shanmukh
         3 hardik jain
```

III. SQL>SELECT*FROMSTUD1;

```
SQL> select *from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

IV. SQL>SELECTSNO,SNAMEFROMSTUD1WHERESNO=1;

```
SQL> select sno,sname from stud1 where sno=1;

       SNO SNAME
---------- -------------
         1 sailaja
```

4) **WRITEAQUERYTOELIMINATEDUPLICATEROWSUSINGSELECT STATEMENT**.

SQL>SELECTDISTINCTSUB1FROMSTUD1;

```
SQL> select distinct sub1 from stud1;

SUB1
--------
engg
Eng
phy
```

5) **WRITEAQUERYTOSORTTHEDATAIN A TABLE**.

SQL>SELECT*FROMSTUD1ORDERBYSNO DESC;

```
SQL> select *from stud1 order by sno desc;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
```

6) **WRITEAQUERYTOCREATETABLEFROM**

ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.

SQL>SELECT*FROMSTUD

```
SQL> select * from stud4;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

7) WRITEAQUERYTOCREATETABLEFROM ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.
SQL>SELECT*FROMSTUD4;

8) WRITEAQUERYTOINSERTDATAINTOATABLEFROMANOTHERTABLE

SQL>CREATETABLESTUD6(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>INSERTINTOSTUD6SELECT*FROMSTUD1WHERE

SNO<=2; 2 ROWS CREATED.

SQL>SELECT*FROMSTUD6;

```
SQL> select * from stud6;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
```

9) WRITEAQUERYTODELETEALLROWS FROM A TABLE.

SQL>DELETEFROMSTUD4;

```
SQL> delete from stud4;

4 rows deleted.
```

10) WRITEAQUERYTOMODIFYTHEEXISTING COLUMNS IN A TABLE.

SQL>SELECTSNAMEASNAMEFROMSTUD1;

```
SQL> select sname as name from stud1;

NAME
--------------------
sailaja
B Samaresh
BS
```

11) WRITEAQUERYTO TRUNCATE A TABLE.

SQL>TRUNCATETABLESTUD1;

TABLE TRUNCATED.

12) WRITEAQUERYTO DESTROY A TABLE.

SQL>DROPTABLESTUD1;

TABLE
DROPPED.

13) WRITEAQUERYTOCREATEATABLECUSTOMERWITHCUSTIDAS PRIMARY KEY FOLLOWED BY OTHER COLUMNS.

SQL>CREATETABLECUSTOMER(CUSTIDNUMBER(4)CONSTRAINTPKLPRIMARY KEY,PRODUCT VARCHAR2(5));

TABLECREATED.

14) WRITEAQUERYTOESTABLISHFOREIGNKEYCONSTRAINT.

SQL> CREATE TABLE PRODUCER (PRODUCT_ID NUMBER (8) UNIQUE,CUST_DEMANDNUMBER(4)REFERENCESCUSTOMER(CUSTID));

TABLECREATED.

15) WRITEAQUERYTOCALCULATETHEANNUALSALARYOFAN EMPLOYEE

SQL>SELECTEMPNO,SAL*12FROMEMP;

```
SQL> select empno,sal*12 from emp;

     EMPNO     SAL*12
---------- ----------
      7839      60000
      7698      34200
      7782      29400
      7566      35700
      7788      36000
      7902      36000
      7369       9600
      7499      19200
      7521      15000
      7654      15000
      7844      18000

     EMPNO     SAL*12
---------- ----------
      7876      13200
      7900      11400
      7934      15600

14 rows selected.
```

16) WRITEAQUERYTOCREATEATABLEHAVINGBNAMEAND BNO, AND DISPLAY VALUE WHERE BNO IS NULL.

SQL>CREATETABLEDEMO1(BNONUMBER(10),BNAME

CHAR(20)); TABLE CREATED.

SQL> INSERT INTO DEMO1
VALUES(1,'MADHU');1ROWCREATED.

SQL> INSERT INTO DEMO1
VALUES(NULL,'SINDHU');1 ROWCREATED.

SQL>INSERTINTODEMO1
VALUES(0,'HEMA');
1ROWCREATED.

SQL>SELECT*FROMDEMO1WHEREBNOISNULL;

```
SQL> select * from demo1 where bno is null;

       BNO BNAME
---------- --------------------
           sindhu
```

17) WRITEAQUERYTOCREATEATABLETHATDOESNOTTAKENULL VALUE.

SQL>CREATETABLEDEMO2(NONUMBER(10),NAMECHAR(20)NOTNULL); TABLE CREATED.

SQL>INSERTINTODEMO2VALUES(1,NULL);

```
SQL> insert into demo2 values(1,null);
insert into demo2 values(1,null)
                          *
ERROR at line 1:
ORA-01400: cannot insert NULL into ("SYSTEM"."DEMO2"."NAME")
```

18) WRITEAQUERYTOIMPLEMENTCONSTRAINTS.

SQL>CREATETABLESTU(SNONUMBER(10)PRIMARYKEY,SNAME CHAR(30) UNIQUE,PHNO NUMBER(10)CHECK(PHNO>800000000),ADDRESSCHAR(10)NOTNULL);

TABLE CREATED.

SQL>DESCSTU;

```
SQL> desc stu;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                       NOT NULL NUMBER(10)
 SNAME                                              CHAR(30)
 PHNO                                               NUMBER(10)
 ADDRESS                                   NOT NULL CHAR(10)
```

19) WRITEAQUERYTODROPPRIMARYKEY CONSTRAINT FROM A TABLE.

SQL>ALTERTABLESTUDROPPRIMARY KEY; TABLE ALTERED.

20) WRITEAQUERYTOIMPLEMENTOR.

SQL>SELECT*FROMEMPWHEREENAME=’FORD’ORSAL=3000;

```
SQL> SELECT * FROM EMP WHERE eNAME='FORD' OR SAL=3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

20) WRITEAQUERYTOIMPLEMENTNOT.

SQL>SELECT*FROMEMPWHERENO SAL>2400;

```
SQL> select * from emp where not SAL>2400;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7369 SMITH      CLERK           7902 17-DEC-80        800
        20

      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300
        30

      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500
        30


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400
        30

      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0
        30

      7876 ADAMS      CLERK           7788 23-MAY-87       1100
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7900 JAMES      CLERK           7698 03-DEC-81        950
        30

      7934 MILLER     CLERK           7782 23-JAN-82       1300
        10
```

21) WRITEAQUERYTOIMPLEMENTRANGE SEARCHING.

SQL>SELECT*FROMEMPWHERESALBETWEEN2000 AND 3000;

```
SQL> select * from emp where sal between 2000 and 3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7782 CLARK      MANAGER         7839 09-JUN-81       2450
        10

      7566 JONES      MANAGER         7839 02-APR-81       2975
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7788 SCOTT      ANALYST         7566 19-APR-87       3000
        20

      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

22) WRITEAQUERYTODISPLAYTHEEMPLOYEENAMETHAT BEGINS WITH Q.

SQL>SELECT*FROMEMPWHEREENAMELIKE'%Q'; NO ROWS SELECTED.

23) WRITEAQUERYTOIMPLEMENTSYSTEM DATE.

SQL>SELECTSYSDATEFROMDUEAL;

```
SQL> SELECT SYSDATE FROM DUAL

SYSDATE
---------
02-OCT-24
```

24) WRITEAQUERYTOFINDTHEAVERAGESALARY OF EMP TABLE

SQL>SELECTAVG(SAL)“AVERAGESALARY”FROMEMP;

```
SQL> SELECT AVG(SAL) “AVERAGE SALARY” FROM EMP;

AVERAGE SALARY
--------------
    2073.21429
```

25) WRITEAQUERYTOFINDTHEMINIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMIN(SAL)“MINIMUMSALARY”FROMEMP;

```
SQL> SELECT MIN(SAL) “MINIMUM SALARY” FROM EMP;

MINIMUM SALARY
--------------
           800
```

26) WRITEAQUERYTOFINDTHEMAXIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMAX(SAL)“MAXIMUMSALARY”FROMEMP;

```
SQL> SELECT MAX(SAL) “MAXIMUM SALARY” FROM EMP;

MAXIMUM SALARY
--------------
          5000
```

27) WRITEAQUERYTOIMPLEMENTCOUNT FUNCTION ON EMP TABLE.

SQL>SELECTCOUNT(*)“TOTALNOOFROWS”FROMEMP;

```
SQL> SELECT COUNT(*) “TOTAL NO OF ROWS” FROM EMP;

TOTAL NO OF ROWS
----------------
              14
```

28) WRITEAQUERYTOIMPLEMENTSUM FUNCTION ON EMP TABLE.

SQL>SELECTSUM(SAL)”SUMOFSLARY”FROMEMP;

```
SQL> SELECT SUM(SAL)”SUM OF SLARY” FROM EMP;

SUM OF SLARY
------------
       29025
```

PL/SQLQUERIES

1. ADD2NUMBERS.

```
SQL>SETSERVEROUTPUTON;
DECLARE
VAR1INTEGER;
VAR2INTEGER;
VAR3INTEGER;
BEGIN
VAR1    :=
&VAR1;VAR2    :=
&VAR2;VAR3:=VAR1+
VAR2;
DBMS_OUTPUT.PUT_LINE('THESUM     OF'||VAR1    ||'AND    '||VAR2    ||    'IS:'||VAR3);
END;
/
```

OUTPUT:

```
Enter value for var1: 10
old   6:   VAR1 := &VAR1;
new   6:   VAR1 := 10;
Enter value for var2: 20
old   7:   VAR2 := &VAR2;
new   7:   VAR2 := 20;
The sum of 10 and 20 is: 30

PL/SQL procedure successfully completed.
```

2. FORLOOPDEMO

```
SQL>SETSERVEROUTPUT
ON;    DECALARE
    INUMBER(2);
    BEGIN
    FORIIN1..10
    LOOP
    DBMS_OUTPUT.PUT_LINE(
    I);    END    LOOP;
    END;
```

OUTPUT:

```
1
2
3
4
5
6
7
8
9
10

PL/SQL procedure successfully completed.
```

3. PL/SQLPROGRAMTOSWAPTWONUMBERS

i. USINGTEMP

```
SQL>SETSERVEROUTPUTON;
DECLARE
 ANUMBER(2):=&A;
BNUMBER(2):=&B;
TEMP
NUMBER(2);
BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE    SWAPPING
 A='||A||'B='||B);    TEMP:=A;
B:=TEMP;
DBMS_OUTPUT.PUT_LINE('AFTER    SWAPPING
 A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 10
old   2:   A NUMBER(2):=&A;
new   2:   A NUMBER(2):=10;
Enter value for b: 20
old   3:   B NUMBER(2):=&B;
new   3:   B NUMBER(2):=20;
BEFORE SWAPPING A=10B=20
AFTER SWAPPING A=10B=10

PL/SQL procedure successfully completed.
```

ii. WITHOUTUSINGTEMP

```
SQL>SETSERVEROUTPUTON;

 DECLARE
1.  NUMBER(2):=&A;
2.  NUMBER
 (2):=&B;    BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE
SWAPPING’A='||A||'B='||B);
A:=A+B;    B:=A-B;
A:=A-B;
DBMS_OUTPUT.PUT_LINE('AFTER
SWAPPING’A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 4
old   2:   A NUMBER(2) := &A;
new   2:   A NUMBER(2) := 4;
Enter value for b: 5
old   3:   B NUMBER(2) := &B;
new   3:   B NUMBER(2) := 5;
BEFORE SWAPPING A=4 B=5
AFTER SWAPPING A=5 B=4

PL/SQL procedure successfully completed.
```

4. WRITEAPROGRAMTOPRINTEMPNO,ENAME,JOB,SALARYOFAN EMPLOYEE OF GIVEN EMPNO.

```
SQL>SETSERVEROUTPUTON;
DECLARE
TYPE EMPREC IS RECORD (ENO
EMP.EMPNO%TYPE,NEMP.ENAME%TYPE,J
EMP.JOB%TYPE,S EMP.SAL%TYPE);
R EMPREC;
ENOEMP.EMPNO%TYPE:=&ENO;
BEGIN
SELECTEMPNO,ENAME,JOB,SAL
INTO R
FROMEMP
WHERE EMPNO = ENO; -- Corrected the
condition
DBMS_OUTPUT.PUT_LINE(R.ENO||''||R.N||''||R.J
||''||R.S);
END;
/
```

OUTPUT:

```
Enter value for eno: 7839
old   4: ENO EMP.EMPNO%TYPE := &ENO;
new   4: ENO EMP.EMPNO%TYPE := 7839;
7839 KING PRESIDENT 5000

PL/SQL procedure successfully completed.
```

5. PL/SQLPROGRAMTOIMPLEMENT%FOUND

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSOREMP_CUR
ISSELECTENAME,COMM
FROMEMPWHEREEMPNO=7654
AND    COMM    IS    NOT    NULL;
EMP_NAME    VARCHAR2(20);
EMP_COMM    NUMBER(10,2);
BEGIN    OPEN    EMP_CUR;
FETCHEMP_CURINTOEMP_NAME,EMP_COMM;
IF    EMP_CUR%FOUND    THEN
DBMS_OUTPUT.PUT_LINE('EMPLOYEE:'||EMP_NAME     ||'GETSCOMMISSION:'||
EMP_COMM);
ELSE
DBMS_OUTPUT.PUT_LINE('EMPLOYEEDIDNOTGETANYCOMMISSION');
END    IF;
DBMS_OUTPUT.PUT_LINE(EMP_CUR%ROWCOUNT||'RECORDSFETCHEDSOFAR');     CLOSE
EMP_CUR;
END;
/
```

OUTPUT

```
EMPLOYEE: MARTIN GETS COMMISSION: 1400
1 RECORDS FETCHED SO FAR

PL/SQL procedure successfully completed.
```

6. PL/SQLTOPMPLEMENT%FOUNDUSINGDEPTTABLE.

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSORDEPTCURISSELECTDEPTNO,DNAMEFROMDEPTWHERE
DEPTNO = 10;
TDEPTCURDEPTCUR%ROWTYPE;
BEGIN
OPENDEPTCUR;
FETCH DEPTCUR INTO
TDEPTCUR; IF
DEPTCUR%FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('DEPTNO:'||TDEPTCUR.DEPTNO||'DNAME:'||
TDEPTCUR.DNAME);
ELSE
DBMS_OUTPUT.PUT_LINE('No department found
withDEPTNO 10'); END IF;
CLOSEDEPTCUR;
END;
/
```

OUTPUT

```
DEPTNO: 10  DNAME: ACCOUNTING

PL/SQL procedure successfully completed.
```

-

7. PL/SQL BLOCK USINGTHE COURSOR TO DISPLAY DETAILS OF ALL EMPLOYEE'SFROMEMPTABLEWHOSESUMOFSALANDCOMMIS >2000

```
SQL>SETSERVEROUTPUTON;
DECLARE
MEM_PNO    EMP.EMPNO%TYPE;
MEM_NAMEEMP.ENAME%TYPE;
MEM_SAL    EMP.SAL%TYPE;
CURSOR    C1    IS
SELECTEMPNO,ENAME,SAL+NVL(COMM,0)
FROM    EMP
WHERESAL+NVL(COMM,0)>2000;
 BEGIN
OPENC1;
DBMS_OUTPUT.PUT_LINE('EMPNO    ENAME    MSAL');
LOOP
FETCHC1INTOMEM_PNO,MEM_NAME,MEM_SAL;
EXIT    WHEN    C1%NOTFOUND;
DBMS_OUTPUT.PUT_LINE(MEM_PNO||"||MEM_NAME     ||"||MEM_SAL);
END    LOOP;
CLOSEC1;
END;
/
```

<u>OUTPUT</u>

```
EMPNO  ENAME   MSAL
7839  KING  5000
7698  BLAKE  2850
7782  CLARK  2450
7566  JONES  2975
7902  FORD  3000
7654  MARTIN  2650

PL/SQL procedure successfully completed.
```

8. **PL/SQL BLOCK THE RETRIVES COURSE TABLE INTO CURSOR.INPUT CODEIDTOSEARCH.IFCOURSEEXISTPRINTTHEINFORMATION.IF COURSE ID DOES NOT EXIST, PRINT USER DEFINED MESSAGE BY THROWING USER DEFINED EXCEPTION.**

```
SQL>SETSERVEROUTPUTON;
DECLARE
  C_NAMECOURSE.COURSENAME%TYPE;
COURSORC1ISSELECTCOURSENAMEFROMCOURSEWHERE
COURSEIS:='&ID';
COURSE_NOT_FOUND
EXCEPTION;

BEGIN
OPENC1;
FETCHC1INTOC_NAME;
IF  C1%  FOUND  THEN
DBMS_OUTPUT.PUT_LINE('COURSE  NAME
IS'||C_NAME);
ELSERAISECOURSE_NOT_FOUND;
END  IF;
EXCEPTIONWHENCOURSE_NOT_FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('COURSE  ID  NOT
FOUND');
END;
/
```

<u>OUTPUT</u>

```
Enter value for id: 13
old 3: cursorcl is select coursename from course where courseis=&id'
new 3:cursor cl is select coursename from course where courseis='13';
course name isB.Sc.computeres
PL/SQL procedure successfully completed.
SQL>/
Enter value for id: 18
old 3: cursor cl is select coursename from course where courseis=&id';
new 3:cursor c1 is select coursename from course where courseis=18';
course id is not found
PL/SQL procedure successfully completed
```

9. PL/SQLPROGRAMTOHANDLEZERODIVIDEEXCEPTION

```
SQL>SETSERVEROUTPUTON;
DECLARE
xNUMBER:=&X;
Y       NUMBER
:=   &Y;   BEGIN
IF   Y   <>   0   THEN
DBMS_OUTPUT.PUT_LINE('DIVISION='||X/
Y);   ELSE
DBMS_OUTPUT.PUT_LINE('CHECK    THE
DENOMINATOR');   END   IF;
END;
 /
```

OUTPUT:

```
Enter value for x: 23
old   2:   X NUMBER := &X;
new   2:   X NUMBER := 23;
Enter value for y: 9
old   3:   Y NUMBER := &Y;
new   3:   Y NUMBER := 9;
DIVISION=2.5555555555555555555555555555555555556

PL/SQL procedure successfully completed.
```

10. PL/SQLTHATCREATESATRIGGERTHATINSERTSORUPDATES THE VALUES OF ENAME.JOB AS LOWER-CASE STRING.

```
SQL> CREATE OR REPLACE TRIGGER
LOWERNAMEBEFOREINSERTORUPDATEONEMP
FOR EACH ROW BEGIN
:NEW.ENAME:=:LOWER(“NEW.ENAME);
NEW.JOB:=LOWER(:MEW.JOB);
END;
/
```

OUTPUT:

Triggercreated.

SQL>UPDATEEEMPSETJOB=’MANAGER’WHEREENAME=SMITH;

```
SQL>  UPDATE EMP SET JOB=’MANAGER’ WHERE ENAME='SMITH';

1 row updated.
```

11. PL/SQLFORSIMPLELOOP.

```
SQL>SETSERVEROUTPUTON;
DECLARE
INUMBER:=0;
BEGIN
LOOP
DBMS_OUTPUT.PUT_LINE('I='||I);
I:=I+2;
EXITWHENI>10;
END LOOP;
END;
/
```

OUTPUT:

```
I=0
I=2
I=4
I=6
I=8
I=10

PL/SQL procedure successfully completed.
```

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During 3rd year 1st semester of the academic year 2025-2026

Name:Mohit Vanka

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Class:DCSIT

Signature of Internal **Signature of HOD**

Signature of External **Signature of Principal**

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STRUCTUREDQUERYLANGUAGE

1) **WRITE A QUERY TO CREATE A TABLE WITH THE FOLLOWINGPARAMETERSSNO,SNAME,DOB,SUB1,SUB2,AND SUB3 USING A APPROPRIATE DATA TYPE.**

SQL>CREATETABLESTUD1(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>DESC

STUD1

```
Table created.

SQL> desc stud1
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                                NUMBER(11)
 SNAME                                     NOT NULL VARCHAR2(13)
 DOB                                                DATE
 SUB1                                               CHAR(8)
 SUB2                                               CHAR(8)
 SUB3                                               CHAR(8)
```

2) WRITEAQUERYTOINSERTDATAINTOTABLEANDDISPLAYTHETABLE.

SQL>INSERTINTOSTUD1VALUES(&SNO,’&SNAME’,’&DOB’,’&SUB1’,’2&SUB2’,’&SUB3’);

```
SQL> insert into stud1 values(&sno,’&sname’,’&dob’,’&sub1’,’2 &sub2’,’&sub3’);
Enter value for sno: 1
Enter value for sname: sailaja
Enter value for dob: 3-oct-1994
Enter value for sub1: Eng
Enter value for sub2: Litera
Enter value for sub3: Phylo
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(1,'sailaja','3-oct-1994','Eng','2 Litera','Phylo')

1 row created.

SQL> /
Enter value for sno: 2
Enter value for sname: shanmukh
Enter value for dob: 11-mar-1999
Enter value for sub1: phy
Enter value for sub2: ldc
Enter value for sub3: m1
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(2,'shanmukh','11-mar-1999','phy','2 ldc','m1')

1 row created.
SQL> /
Enter value for sno: 3
Enter value for sname: hardik jain
Enter value for dob: 02-oct-2005
Enter value for sub1: engg
Enter value for sub2: m2
Enter value for sub3: pis
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(3,'hardik jain','02-oct-2005','engg','2 m2','pis')

1 row created.
```

SQL>SELECT*FROMSTUD1;

```
SQL> select * from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

3) WRITEAQUERYTODISPLAYTHEDATAINTABLEASI.SELECTROWSAND COLUMNS II. SELECTED COLUMNS AND ALL ROWS ALL ROWS III. ALL ROWS AND ALL COLUMNS IV. SELECTED ROWS AND ALL COLUMNS

I. SQL>SELECT*FROMSTUD1WHERESNO>2;

```
SQL> select * from stud1 where sno>2;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

II. SQL>SELECTSNO,SNAMEFROMSTUD1;

```
SQL> select sno,sname from stud1;

       SNO SNAME
---------- -------------
         1 sailaja
         2 shanmukh
         3 hardik jain
```

III. SQL>SELECT*FROMSTUD1;

```
SQL> select *from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

IV. SQL>SELECTSNO,SNAMEFROMSTUD1WHERESNO=1;

```
SQL> select sno,sname from stud1 where sno=1;

       SNO SNAME
---------- -------------
         1 sailaja
```

4) **WRITEAQUERYTOELIMINATEDUPLICATEROWSUSINGSELECT STATEMENT**.

SQL>SELECTDISTINCTSUB1FROMSTUD1;

```
SQL> select distinct sub1 from stud1;

SUB1
--------
engg
Eng
phy
```

5) **WRITEAQUERYTOSORTTHEDATAIN A TABLE**.

SQL>SELECT*FROMSTUD1ORDERBYSNO DESC;

```
SQL> select *from stud1 order by sno desc;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
```

6) **WRITEAQUERYTOCREATETABLEFROM**

ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.

SQL>SELECT*FROMSTUD

```
SQL> select * from stud4;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

7) WRITEAQUERYTOCREATETABLEFROM ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.
SQL>SELECT*FROMSTUD4;

8) WRITEAQUERYTOINSERTDATAINTOATABLEFROMANOTHERTABLE

SQL>CREATETABLESTUD6(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>INSERTINTOSTUD6SELECT*FROMSTUD1WHERE

SNO<=2; 2 ROWS CREATED.

SQL>SELECT*FROMSTUD6;

```
SQL> select * from stud6;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
```

9) WRITEAQUERYTODELETEALLROWS FROM A TABLE.

SQL>DELETEFROMSTUD4;

```
SQL> delete from stud4;

4 rows deleted.
```

10) WRITEAQUERYTOMODIFYTHEEXISTING COLUMNS IN A TABLE.

SQL>SELECTSNAMEASNAMEFROMSTUD1;

```
SQL> select sname as name from stud1;

NAME
--------------------
sailaja
B Samaresh
BS
```

11) WRITEAQUERYTO TRUNCATE A TABLE.

SQL>TRUNCATETABLESTUD1;

TABLE TRUNCATED.

12) WRITEAQUERYTO DESTROY A TABLE.

SQL>DROPTABLESTUD1;

TABLE
DROPPED.

13) WRITEAQUERYTOCREATEATABLECUSTOMERWITHCUSTIDAS PRIMARY KEY FOLLOWED BY OTHER COLUMNS.

SQL>CREATETABLECUSTOMER(CUSTIDNUMBER(4)CONSTRAINTPKLPRIMARY KEY,PRODUCT VARCHAR2(5));

TABLECREATED.

14) WRITEAQUERYTOESTABLISHFOREIGNKEYCONSTRAINT.

SQL> CREATE TABLE PRODUCER (PRODUCT_ID NUMBER (8) UNIQUE,CUST_DEMANDNUMBER(4)REFERENCESCUSTOMER(CUSTID));

TABLECREATED.

15) WRITEAQUERYTOCALCULATETHEANNUALSALARYOFAN EMPLOYEE

SQL>SELECTEMPNO,SAL*12FROMEMP;

```
SQL> select empno,sal*12 from emp;

     EMPNO     SAL*12
---------- ----------
      7839      60000
      7698      34200
      7782      29400
      7566      35700
      7788      36000
      7902      36000
      7369       9600
      7499      19200
      7521      15000
      7654      15000
      7844      18000

     EMPNO     SAL*12
---------- ----------
      7876      13200
      7900      11400
      7934      15600

14 rows selected.
```

16) WRITEAQUERYTOCREATEATABLEHAVINGBNAMEAND BNO, AND DISPLAY VALUE WHERE BNO IS NULL.

SQL>CREATETABLEDEMO1(BNONUMBER(10),BNAME

CHAR(20)); TABLE CREATED.

SQL> INSERT INTO DEMO1
VALUES(1,'MADHU');1ROWCREATED.

SQL> INSERT INTO DEMO1
VALUES(NULL,'SINDHU');1 ROWCREATED.

SQL>INSERTINTODEMO1
VALUES(0,'HEMA');
1ROWCREATED.

SQL>SELECT*FROMDEMO1WHEREBNOISNULL;

```
SQL> select * from demo1 where bno is null;

       BNO BNAME
---------- --------------------
           sindhu
```

17) WRITEAQUERYTOCREATEATABLETHATDOESNOTTAKENULL VALUE.

SQL>CREATETABLEDEMO2(NONUMBER(10),NAMECHAR(20)NOTNULL); TABLE CREATED.

SQL>INSERTINTODEMO2VALUES(1,NULL);

```
SQL> insert into demo2 values(1,null);
insert into demo2 values(1,null)
                          *
ERROR at line 1:
ORA-01400: cannot insert NULL into ("SYSTEM"."DEMO2"."NAME")
```

18) WRITEAQUERYTOIMPLEMENTCONSTRAINTS.

SQL>CREATETABLESTU(SNONUMBER(10)PRIMARYKEY,SNAME CHAR(30) UNIQUE,PHNO NUMBER(10)CHECK(PHNO>800000000),ADDRESSCHAR(10)NOTNULL);

TABLE CREATED.

SQL>DESCSTU;

```
SQL> desc stu;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                       NOT NULL NUMBER(10)
 SNAME                                              CHAR(30)
 PHNO                                               NUMBER(10)
 ADDRESS                                   NOT NULL CHAR(10)
```

19) WRITEAQUERYTODROPPRIMARYKEY CONSTRAINT FROM A TABLE.

SQL>ALTERTABLESTUDROPPRIMARY KEY; TABLE ALTERED.

20) WRITEAQUERYTOIMPLEMENTOR.

SQL>SELECT*FROMEMPWHEREENAME=’FORD’ORSAL=3000;

```
SQL> SELECT * FROM EMP WHERE eNAME='FORD' OR SAL=3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

20) WRITEAQUERYTOIMPLEMENTNOT.

SQL>SELECT*FROMEMPWHERENO SAL>2400;

```
SQL> select * from emp where not SAL>2400;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7369 SMITH      CLERK           7902 17-DEC-80        800
        20

      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300
        30

      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500
        30


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400
        30

      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0
        30

      7876 ADAMS      CLERK           7788 23-MAY-87       1100
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7900 JAMES      CLERK           7698 03-DEC-81        950
        30

      7934 MILLER     CLERK           7782 23-JAN-82       1300
        10
```

21) WRITEAQUERYTOIMPLEMENTRANGE SEARCHING.

SQL>SELECT*FROMEMPWHERESALBETWEEN2000 AND 3000;

```
SQL> select * from emp where sal between 2000 and 3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7782 CLARK      MANAGER         7839 09-JUN-81       2450
        10

      7566 JONES      MANAGER         7839 02-APR-81       2975
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7788 SCOTT      ANALYST         7566 19-APR-87       3000
        20

      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

22) WRITEAQUERYTODISPLAYTHEEMPLOYEENAMETHAT BEGINS WITH Q.

SQL>SELECT*FROMEMPWHEREENAMELIKE'%Q'; NO ROWS SELECTED.

23) WRITEAQUERYTOIMPLEMENTSYSTEM DATE.

SQL>SELECTSYSDATEFROMDUEAL;

```
SQL> SELECT SYSDATE FROM DUAL

SYSDATE
---------
02-OCT-24
```

24) WRITEAQUERYTOFINDTHEAVERAGESALARY OF EMP TABLE

SQL>SELECTAVG(SAL)“AVERAGESALARY”FROMEMP;

```
SQL> SELECT AVG(SAL) “AVERAGE SALARY” FROM EMP;

AVERAGE SALARY
--------------
    2073.21429
```

25) WRITEAQUERYTOFINDTHEMINIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMIN(SAL)“MINIMUMSALARY”FROMEMP;

```
SQL> SELECT MIN(SAL) “MINIMUM SALARY” FROM EMP;

MINIMUM SALARY
--------------
           800
```

26) WRITEAQUERYTOFINDTHEMAXIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMAX(SAL)“MAXIMUMSALARY”FROMEMP;

```
SQL> SELECT MAX(SAL) “MAXIMUM SALARY” FROM EMP;

MAXIMUM SALARY
--------------
          5000
```

27) WRITEAQUERYTOIMPLEMENTCOUNT FUNCTION ON EMP TABLE.

SQL>SELECTCOUNT(*)“TOTALNOOFROWS”FROMEMP;

```
SQL> SELECT COUNT(*) “TOTAL NO OF ROWS” FROM EMP;

TOTAL NO OF ROWS
----------------
              14
```

28) WRITEAQUERYTOIMPLEMENTSUM FUNCTION ON EMP TABLE.

SQL>SELECTSUM(SAL)”SUMOFSLARY”FROMEMP;

```
SQL> SELECT SUM(SAL)”SUM OF SLARY” FROM EMP;

SUM OF SLARY
------------
       29025
```

PL/SQLQUERIES

1. ADD2NUMBERS.

```
SQL>SETSERVEROUTPUTON;
DECLARE
VAR1INTEGER;
VAR2INTEGER;
VAR3INTEGER;
BEGIN
VAR1    :=
&VAR1;VAR2    :=
&VAR2;VAR3:=VAR1+
VAR2;
DBMS_OUTPUT.PUT_LINE('THESUM     OF'||VAR1    ||'AND    '||VAR2    ||    'IS:'||VAR3);
END;
/
```

OUTPUT:

```
Enter value for var1: 10
old   6:   VAR1 := &VAR1;
new   6:   VAR1 := 10;
Enter value for var2: 20
old   7:   VAR2 := &VAR2;
new   7:   VAR2 := 20;
The sum of 10 and 20 is: 30

PL/SQL procedure successfully completed.
```

2. FORLOOPDEMO

```
SQL>SETSERVEROUTPUT
ON;    DECALARE
    INUMBER(2);
    BEGIN
    FORIIN1..10
    LOOP
    DBMS_OUTPUT.PUT_LINE(
    I);    END    LOOP;
    END;
```

OUTPUT:

```
1
2
3
4
5
6
7
8
9
10

PL/SQL procedure successfully completed.
```

3. PL/SQLPROGRAMTOSWAPTWONUMBERS

i. USINGTEMP

```
SQL>SETSERVEROUTPUTON;
DECLARE
 ANUMBER(2):=&A;
BNUMBER(2):=&B;
TEMP
NUMBER(2);
BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE    SWAPPING
 A='||A||'B='||B);    TEMP:=A;
B:=TEMP;
DBMS_OUTPUT.PUT_LINE('AFTER    SWAPPING
 A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 10
old   2:   A NUMBER(2):=&A;
new   2:   A NUMBER(2):=10;
Enter value for b: 20
old   3:   B NUMBER(2):=&B;
new   3:   B NUMBER(2):=20;
BEFORE SWAPPING A=10B=20
AFTER SWAPPING A=10B=10

PL/SQL procedure successfully completed.
```

ii. WITHOUTUSINGTEMP

```
SQL>SETSERVEROUTPUTON;

 DECLARE
1.  NUMBER(2):=&A;
2.  NUMBER
 (2):=&B;    BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE
SWAPPING’A='||A||'B='||B);
A:=A+B;    B:=A-B;
A:=A-B;
DBMS_OUTPUT.PUT_LINE('AFTER
SWAPPING’A='||A||'B='||B);     END;
/
```

OUTPUT:

```
Enter value for a: 4
old   2:   A NUMBER(2) := &A;
new   2:   A NUMBER(2) := 4;
Enter value for b: 5
old   3:   B NUMBER(2) := &B;
new   3:   B NUMBER(2) := 5;
BEFORE SWAPPING A=4 B=5
AFTER SWAPPING A=5 B=4

PL/SQL procedure successfully completed.
```

4. WRITEAPROGRAMTOPRINTEMPNO,ENAME,JOB,SALARYOFAN EMPLOYEE OF GIVEN EMPNO.

```
SQL>SETSERVEROUTPUTON;

DECLARE

TYPE EMPREC IS RECORD (ENO
EMP.EMPNO%TYPE,NEMP.ENAME%TYPE,J
EMP.JOB%TYPE,S EMP.SAL%TYPE);

R EMPREC;

ENOEMP.EMPNO%TYPE:=&ENO;

BEGIN

SELECTEMPNO,ENAME,JOB,SAL

INTO R

FROMEMP
WHERE EMPNO = ENO; -- Corrected the
condition
DBMS_OUTPUT.PUT_LINE(R.ENO||''||R.N||''||R.J
||''||R.S);
END;
/
```

OUTPUT:

```
Enter value for eno: 7839
old   4: ENO EMP.EMPNO%TYPE := &ENO;
new   4: ENO EMP.EMPNO%TYPE := 7839;
7839 KING PRESIDENT 5000

PL/SQL procedure successfully completed.
```

5. PL/SQLPROGRAMTOIMPLEMENT%FOUND

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSOREMP_CUR
ISSELECTENAME,COMM
FROMEMPWHEREEMPNO=7654
AND    COMM    IS    NOT    NULL;
EMP_NAME    VARCHAR2(20);
EMP_COMM    NUMBER(10,2);
BEGIN    OPEN    EMP_CUR;
FETCHEMP_CURINTOEMP_NAME,EMP_COMM;
IF    EMP_CUR%FOUND    THEN
DBMS_OUTPUT.PUT_LINE('EMPLOYEE:'||EMP_NAME     ||'GETSCOMMISSION:'||
EMP_COMM);
ELSE
DBMS_OUTPUT.PUT_LINE('EMPLOYEEDIDNOTGETANYCOMMISSION');
END    IF;
DBMS_OUTPUT.PUT_LINE(EMP_CUR%ROWCOUNT||'RECORDSFETCHEDSOFAR');     CLOSE
EMP_CUR;
END;
/
```

OUTPUT

```
EMPLOYEE: MARTIN GETS COMMISSION: 1400
1 RECORDS FETCHED SO FAR

PL/SQL procedure successfully completed.
```

6. PL/SQLTOPMPLEMENT%FOUNDUSINGDEPTTABLE.

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSORDEPTCURISSELECTDEPTNO,DNAMEFROMDEPTWHERE
DEPTNO = 10;
TDEPTCURDEPTCUR%ROWTYPE;
BEGIN
OPENDEPTCUR;
FETCH DEPTCUR INTO
TDEPTCUR; IF
DEPTCUR%FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('DEPTNO:'||TDEPTCUR.DEPTNO||'DNAME:'||
TDEPTCUR.DNAME);
ELSE
DBMS_OUTPUT.PUT_LINE('No department found
withDEPTNO 10'); END IF;
CLOSEDEPTCUR;
END;
/
```

OUTPUT

```
DEPTNO: 10  DNAME: ACCOUNTING

PL/SQL procedure successfully completed.
```

-

7. PL/SQL BLOCK USINGTHE COURSOR TO DISPLAY DETAILS OF ALL EMPLOYEE'SFROMEMPTABLEWHOSESUMOFSALANDCOMMIS >2000

```
SQL>SETSERVEROUTPUTON;
DECLARE
MEM_PNO    EMP.EMPNO%TYPE;
MEM_NAMEEMP.ENAME%TYPE;
MEM_SAL    EMP.SAL%TYPE;
CURSOR    C1    IS
SELECTEMPNO,ENAME,SAL+NVL(COMM,0)
FROM    EMP
WHERESAL+NVL(COMM,0)>2000;
 BEGIN
OPENC1;
DBMS_OUTPUT.PUT_LINE('EMPNO    ENAME    MSAL');
LOOP
FETCHC1INTOMEM_PNO,MEM_NAME,MEM_SAL;
EXIT    WHEN    C1%NOTFOUND;
DBMS_OUTPUT.PUT_LINE(MEM_PNO||"||MEM_NAME     ||"||MEM_SAL);
END    LOOP;
CLOSEC1;
END;
/
```

OUTPUT

```
EMPNO  ENAME   MSAL
7839  KING  5000
7698  BLAKE  2850
7782  CLARK  2450
7566  JONES  2975
7902  FORD  3000
7654  MARTIN  2650

PL/SQL procedure successfully completed.
```

8. **PL/SQL BLOCK THE RETRIVES COURSE TABLE INTO CURSOR.INPUT CODEIDTOSEARCH.IFCOURSEEXISTPRINTTHEINFORMATION.IF COURSE ID DOES NOT EXIST, PRINT USER DEFINED MESSAGE BY THROWING USER DEFINED EXCEPTION.**

```
SQL>SETSERVEROUTPUTON;
DECLARE
  C_NAMECOURSE.COURSENAME%TYPE;
COURSORC1ISSELECTCOURSENAMEFROMCOURSEWHERE
COURSEIS:='&ID';
COURSE_NOT_FOUND
EXCEPTION;

BEGIN
OPENC1;
FETCHC1INTOC_NAME;
IF C1% FOUND THEN
DBMS_OUTPUT.PUT_LINE('COURSE NAME
IS'||C_NAME);
ELSERAISECOURSE_NOT_FOUND;
END IF;
EXCEPTIONWHENCOURSE_NOT_FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('COURSE ID NOT
FOUND');
END;
/
```

<u>OUTPUT</u>

```
Enter value for id: 13
old 3: cursorcl is select coursename from course where courseis=&id'
new 3:cursor cl is select coursename from course where courseis='13';
course name isB.Sc.computeres
PL/SQL procedure successfully completed.
SQL>/
Enter value for id: 18
old 3: cursor cl is select coursename from course where courseis=&id';
new 3:cursor c1 is select coursename from course where courseis=18';
course id is not found
PL/SQL procedure successfully completed
```

9. PL/SQLPROGRAMTOHANDLEZERODIVIDEEXCEPTION

```
SQL>SETSERVEROUTPUTON;
DECLARE
xNUMBER:=&X;
Y       NUMBER
:=   &Y;   BEGIN
IF   Y   <>   0   THEN
DBMS_OUTPUT.PUT_LINE('DIVISION='||X/
Y);   ELSE
DBMS_OUTPUT.PUT_LINE('CHECK    THE
DENOMINATOR');   END   IF;
END;
 /
```

OUTPUT:

```
Enter value for x: 23
old   2:   X NUMBER := &X;
new   2:   X NUMBER := 23;
Enter value for y: 9
old   3:   Y NUMBER := &Y;
new   3:   Y NUMBER := 9;
DIVISION=2.5555555555555555555555555555555555556

PL/SQL procedure successfully completed.
```

10. PL/SQLTHATCREATESATRIGGERTHATINSERTSORUPDATES THE VALUES OF ENAME.JOB AS LOWER-CASE STRING.

```
SQL> CREATE OR REPLACE TRIGGER
LOWERNAMEBEFOREINSERTORUPDATEONEMP
FOR EACH ROW BEGIN
:NEW.ENAME:=:LOWER(“NEW.ENAME);
NEW.JOB:=LOWER(:MEW.JOB);
END;
/
```

OUTPUT:

Triggercreated.

SQL>UPDATEEEMPSETJOB=’MANAGER’WHEREENAME=SMITH;

```
SQL>  UPDATE EMP SET JOB=’MANAGER’ WHERE ENAME='SMITH';

1 row updated.
```

11. PL/SQLFORSIMPLELOOP.

```
SQL>SETSERVEROUTPUTON;
DECLARE
INUMBER:=0;
BEGIN
LOOP
DBMS_OUTPUT.PUT_LINE('I='||I);
I:=I+2;
EXITWHENI>10;
END LOOP;
END;
/
```

OUTPUT:

```
I=0
I=2
I=4
I=6
I=8
I=10

PL/SQL procedure successfully completed.
```

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During 3rd year 1st semester of the academic year 2025-2026

Name: BUTTI NISHANK RAJ

UIDNo:111723016014

Class:DCSIT

Signature of Internal **Signature of HOD**

Signature of External **Signature of Principal**

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STRUCTUREDQUERYLANGUAGE

1) **WRITE A QUERY TO CREATE A TABLE WITH THE FOLLOWINGPARAMETERSSNO,SNAME,DOB,SUB1,SUB2,AND SUB3 USING A APPROPRIATE DATA TYPE.**

SQL>CREATETABLESTUD1(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>DESC

STUD1

```
Table created.

SQL> desc stud1
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                                NUMBER(11)
 SNAME                                     NOT NULL VARCHAR2(13)
 DOB                                                DATE
 SUB1                                               CHAR(8)
 SUB2                                               CHAR(8)
 SUB3                                               CHAR(8)
```

2) WRITEAQUERYTOINSERTDATAINTOTABLEANDDISPLAYTHETABLE.

SQL>INSERTINTOSTUD1VALUES(&SNO,’&SNAME’,’&DOB’,’&SUB1’,’2&SUB2’,’&SUB3’);

```
SQL> insert into stud1 values(&sno,’&sname’,’&dob’,’&sub1’,’2 &sub2’,’&sub3’);
Enter value for sno: 1
Enter value for sname: sailaja
Enter value for dob: 3-oct-1994
Enter value for sub1: Eng
Enter value for sub2: Litera
Enter value for sub3: Phylo
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(1,'sailaja','3-oct-1994','Eng','2 Litera','Phylo')

1 row created.

SQL> /
Enter value for sno: 2
Enter value for sname: shanmukh
Enter value for dob: 11-mar-1999
Enter value for sub1: phy
Enter value for sub2: ldc
Enter value for sub3: m1
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(2,'shanmukh','11-mar-1999','phy','2 ldc','m1')

1 row created.
SQL> /
Enter value for sno: 3
Enter value for sname: hardik jain
Enter value for dob: 02-oct-2005
Enter value for sub1: engg
Enter value for sub2: m2
Enter value for sub3: pis
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(3,'hardik jain','02-oct-2005','engg','2 m2','pis')

1 row created.
```

SQL>SELECT*FROMSTUD1;

```
SQL> select * from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

3) **WRITEAQUERYTODISPLAYTHEDATAINTABLEASI.SELECTROWSAND COLUMNS II. SELECTED COLUMNS AND ALL ROWS ALL ROWS III. ALL ROWS AND ALL COLUMNS IV. SELECTED ROWS AND ALL COLUMNS**

I. SQL>SELECT*FROMSTUD1WHERESNO>2;

```
SQL> select * from stud1 where sno>2;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

II. SQL>SELECTSNO,SNAMEFROMSTUD1;

```
SQL> select sno,sname from stud1;

       SNO SNAME
---------- -------------
         1 sailaja
         2 shanmukh
         3 hardik jain
```

III. SQL>SELECT*FROMSTUD1;

```
SQL> select *from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

IV. SQL>SELECTSNO,SNAMEFROMSTUD1WHERESNO=1;

```
SQL> select sno,sname from stud1 where sno=1;

       SNO SNAME
---------- -------------
         1 sailaja
```

4) **WRITEAQUERYTOELIMINATEDUPLICATEROWSUSINGSELECT STATEMENT**.

SQL>SELECTDISTINCTSUB1FROMSTUD1;

```
SQL> select distinct sub1 from stud1;

SUB1
--------
engg
Eng
phy
```

5) **WRITEAQUERYTOSORTTHEDATAIN A TABLE**.

SQL>SELECT*FROMSTUD1ORDERBYSNO DESC;

```
SQL> select *from stud1 order by sno desc;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
```

6) **WRITEAQUERYTOCREATETABLEFROM**

ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.

SQL>SELECT*FROMSTUD

```
SQL> select * from stud4;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

7) WRITEAQUERYTOCREATETABLEFROM ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.
SQL>SELECT*FROMSTUD4;

8) WRITEAQUERYTOINSERTDATAINTOATABLEFROMANOTHERTABLE

SQL>CREATETABLESTUD6(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>INSERTINTOSTUD6SELECT*FROMSTUD1WHERE

SNO<=2; 2 ROWS CREATED.

SQL>SELECT*FROMSTUD6;

```
SQL> select * from stud6;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
```

9) WRITEAQUERYTODELETEALLROWS FROM A TABLE.

SQL>DELETEFROMSTUD4;

```
SQL> delete from stud4;

4 rows deleted.
```

10) WRITEAQUERYTOMODIFYTHEEXISTING COLUMNS IN A TABLE.

SQL>SELECTSNAMEASNAMEFROMSTUD1;

```
SQL> select sname as name from stud1;

NAME
--------------------
sailaja
B Samaresh
BS
```

11) WRITEAQUERYTO TRUNCATE A TABLE.

SQL>TRUNCATETABLESTUD1;

TABLE TRUNCATED.

12) WRITEAQUERYTO DESTROY A TABLE.

SQL>DROPTABLESTUD1;

TABLE
DROPPED.

13) WRITEAQUERYTOCREATEATABLECUSTOMERWITHCUSTIDAS PRIMARY KEY FOLLOWED BY OTHER COLUMNS.

SQL>CREATETABLECUSTOMER(CUSTIDNUMBER(4)CONSTRAINTPKLPRIMARY KEY,PRODUCT VARCHAR2(5));

TABLECREATED.

14) WRITEAQUERYTOESTABLISHFOREIGNKEYCONSTRAINT.

SQL> CREATE TABLE PRODUCER (PRODUCT_ID NUMBER (8) UNIQUE,CUST_DEMANDNUMBER(4)REFERENCESCUSTOMER(CUSTID));

TABLECREATED.

15) WRITEAQUERYTOCALCULATETHEANNUALSALARYOFAN EMPLOYEE

SQL>SELECTEMPNO,SAL*12FROMEMP;

```
SQL> select empno,sal*12 from emp;

     EMPNO     SAL*12
---------- ----------
      7839      60000
      7698      34200
      7782      29400
      7566      35700
      7788      36000
      7902      36000
      7369       9600
      7499      19200
      7521      15000
      7654      15000
      7844      18000

     EMPNO     SAL*12
---------- ----------
      7876      13200
      7900      11400
      7934      15600

14 rows selected.
```

16) WRITEAQUERYTOCREATEATABLEHAVINGBNAMEAND BNO, AND DISPLAY VALUE WHERE BNO IS NULL.

SQL>CREATETABLEDEMO1(BNONUMBER(10),BNAME

CHAR(20)); TABLE CREATED.

SQL> INSERT INTO DEMO1
VALUES(1,'MADHU');1ROWCREATED.

SQL> INSERT INTO DEMO1
VALUES(NULL,'SINDHU');1 ROWCREATED.

SQL>INSERTINTODEMO1
VALUES(0,'HEMA');
1ROWCREATED.

SQL>SELECT*FROMDEMO1WHEREBNOISNULL;

```
SQL> select * from demo1 where bno is null;

       BNO BNAME
---------- --------------------
           sindhu
```

17) WRITEAQUERYTOCREATEATABLETHATDOESNOTTAKENULL VALUE.

SQL>CREATETABLEDEMO2(NONUMBER(10),NAMECHAR(20)NOTNULL); TABLE CREATED.

SQL>INSERTINTODEMO2VALUES(1,NULL);

```
SQL> insert into demo2 values(1,null);
insert into demo2 values(1,null)
                          *
ERROR at line 1:
ORA-01400: cannot insert NULL into ("SYSTEM"."DEMO2"."NAME")
```

18) WRITEAQUERYTOIMPLEMENTCONSTRAINTS.

SQL>CREATETABLESTU(SNONUMBER(10)PRIMARYKEY,SNAME CHAR(30) UNIQUE,PHNO NUMBER(10)CHECK(PHNO>800000000),ADDRESSCHAR(10)NOTNULL);

TABLE CREATED.

SQL>DESCSTU;

```
SQL> desc stu;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                       NOT NULL NUMBER(10)
 SNAME                                              CHAR(30)
 PHNO                                               NUMBER(10)
 ADDRESS                                   NOT NULL CHAR(10)
```

19) WRITEAQUERYTODROPPRIMARYKEY CONSTRAINT FROM A TABLE.

SQL>ALTERTABLESTUDROPPRIMARY KEY; TABLE ALTERED.

20) WRITEAQUERYTOIMPLEMENTOR.

SQL>SELECT*FROMEMPWHEREENAME=’FORD’ORSAL=3000;

```
SQL> SELECT * FROM EMP WHERE eNAME='FORD' OR SAL=3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

20) WRITEAQUERYTOIMPLEMENTNOT.

SQL>SELECT*FROMEMPWHERENO SAL>2400;

```
SQL> select * from emp where not SAL>2400;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7369 SMITH      CLERK           7902 17-DEC-80        800
        20

      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300
        30

      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500
        30


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400
        30

      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0
        30

      7876 ADAMS      CLERK           7788 23-MAY-87       1100
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7900 JAMES      CLERK           7698 03-DEC-81        950
        30

      7934 MILLER     CLERK           7782 23-JAN-82       1300
        10
```

21) WRITEAQUERYTOIMPLEMENTRANGE SEARCHING.

SQL>SELECT*FROMEMPWHERESALBETWEEN2000 AND 3000;

```
SQL> select * from emp where sal between 2000 and 3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7782 CLARK      MANAGER         7839 09-JUN-81       2450
        10

      7566 JONES      MANAGER         7839 02-APR-81       2975
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7788 SCOTT      ANALYST         7566 19-APR-87       3000
        20

      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

22) WRITEAQUERYTODISPLAYTHEEMPLOYEENAMETHAT BEGINS WITH Q.

SQL>SELECT*FROMEMPWHEREENAMELIKE'%Q'; NO ROWS SELECTED.

23) WRITEAQUERYTOIMPLEMENTSYSTEM DATE.

SQL>SELECTSYSDATEFROMDUEAL;

```
SQL> SELECT SYSDATE FROM DUAL

SYSDATE
---------
02-OCT-24
```

24) WRITEAQUERYTOFINDTHEAVERAGESALARY OF EMP TABLE

SQL>SELECTAVG(SAL)“AVERAGESALARY”FROMEMP;

```
SQL> SELECT AVG(SAL) “AVERAGE SALARY” FROM EMP;

AVERAGE SALARY
--------------
    2073.21429
```

25) WRITEAQUERYTOFINDTHEMINIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMIN(SAL)“MINIMUMSALARY”FROMEMP;

```
SQL> SELECT MIN(SAL) “MINIMUM SALARY” FROM EMP;

MINIMUM SALARY
--------------
           800
```

26) WRITEAQUERYTOFINDTHEMAXIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMAX(SAL)“MAXIMUMSALARY”FROMEMP;

```
SQL> SELECT MAX(SAL) “MAXIMUM SALARY” FROM EMP;

MAXIMUM SALARY
--------------
          5000
```

27) WRITEAQUERYTOIMPLEMENTCOUNT FUNCTION ON EMP TABLE.

SQL>SELECTCOUNT(*)“TOTALNOOFROWS”FROMEMP;

```
SQL> SELECT COUNT(*) “TOTAL NO OF ROWS” FROM EMP;

TOTAL NO OF ROWS
----------------
              14
```

28) WRITEAQUERYTOIMPLEMENTSUM FUNCTION ON EMP TABLE.

SQL>SELECTSUM(SAL)”SUMOFSLARY”FROMEMP;

```
SQL> SELECT SUM(SAL)”SUM OF SLARY” FROM EMP;

SUM OF SLARY
------------
       29025
```

PL/SQLQUERIES

1. ADD2NUMBERS.

```
SQL>SETSERVEROUTPUTON;
DECLARE
VAR1INTEGER;
VAR2INTEGER;
VAR3INTEGER;
BEGIN
VAR1    :=
&VAR1;VAR2    :=
&VAR2;VAR3:=VAR1+
VAR2;
DBMS_OUTPUT.PUT_LINE('THESUM    OF'||VAR1    ||'AND    '||VAR2    ||    'IS:'||VAR3);
END;
/
```

OUTPUT:

```
Enter value for var1: 10
old   6:   VAR1 := &VAR1;
new   6:   VAR1 := 10;
Enter value for var2: 20
old   7:   VAR2 := &VAR2;
new   7:   VAR2 := 20;
The sum of 10 and 20 is: 30

PL/SQL procedure successfully completed.
```

2. FORLOOPDEMO

```
SQL>SETSERVEROUTPUT
ON;    DECALARE
    INUMBER(2);
    BEGIN
    FORIIN1..10
    LOOP
    DBMS_OUTPUT.PUT_LINE(
    I);    END    LOOP;
    END;
```

OUTPUT:

```
1
2
3
4
5
6
7
8
9
10

PL/SQL procedure successfully completed.
```

3. PL/SQLPROGRAMTOSWAPTWONUMBERS

i. USINGTEMP

```
SQL>SETSERVEROUTPUTON;
DECLARE
 ANUMBER(2):=&A;
BNUMBER(2):=&B;
TEMP
NUMBER(2);
BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE    SWAPPING
 A='||A||'B='||B);    TEMP:=A;
B:=TEMP;
DBMS_OUTPUT.PUT_LINE('AFTER    SWAPPING
 A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 10
old   2:   A NUMBER(2):=&A;
new   2:   A NUMBER(2):=10;
Enter value for b: 20
old   3:   B NUMBER(2):=&B;
new   3:   B NUMBER(2):=20;
BEFORE SWAPPING A=10B=20
AFTER SWAPPING A=10B=10

PL/SQL procedure successfully completed.
```

ii. WITHOUTUSINGTEMP

```
SQL>SETSERVEROUTPUTON;
 DECLARE
1.  NUMBER(2):=&A;
2.  NUMBER
 (2):=&B;    BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE
SWAPPING’A='||A||'B='||B);
A:=A+B;    B:=A-B;
A:=A-B;
DBMS_OUTPUT.PUT_LINE('AFTER
SWAPPING’A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 4
old   2:   A NUMBER(2) := &A;
new   2:   A NUMBER(2) := 4;
Enter value for b: 5
old   3:   B NUMBER(2) := &B;
new   3:   B NUMBER(2) := 5;
BEFORE SWAPPING A=4 B=5
AFTER SWAPPING A=5 B=4

PL/SQL procedure successfully completed.
```

4. WRITEAPROGRAMTOPRINTEMPNO,ENAME,JOB,SALARYOFAN EMPLOYEE OF GIVEN EMPNO.

```
SQL>SETSERVEROUTPUTON;

DECLARE

TYPE EMPREC IS RECORD (ENO
EMP.EMPNO%TYPE,NEMP.ENAME%TYPE,J
EMP.JOB%TYPE,S EMP.SAL%TYPE);

R EMPREC;

ENOEMP.EMPNO%TYPE:=&ENO;

BEGIN

SELECTEMPNO,ENAME,JOB,SAL

INTO R

FROMEMP
WHERE EMPNO = ENO; -- Corrected the
condition

DBMS_OUTPUT.PUT_LINE(R.ENO||''||R.N||''||R.J
||''||R.S);

END;
/
```

OUTPUT:

```
Enter value for eno: 7839
old   4: ENO EMP.EMPNO%TYPE := &ENO;
new   4: ENO EMP.EMPNO%TYPE := 7839;
7839 KING PRESIDENT 5000

PL/SQL procedure successfully completed.
```

5. PL/SQLPROGRAMTOIMPLEMENT%FOUND

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSOREMP_CUR
ISSELECTENAME,COMM
FROMEMPWHEREEMPNO=7654
AND    COMM    IS    NOT    NULL;
EMP_NAME    VARCHAR2(20);
EMP_COMM    NUMBER(10,2);
BEGIN    OPEN    EMP_CUR;
FETCHEMP_CURINTOEMP_NAME,EMP_COMM;
IF    EMP_CUR%FOUND    THEN
DBMS_OUTPUT.PUT_LINE('EMPLOYEE:'||EMP_NAME         ||'GETSCOMMISSION:'||
EMP_COMM);
ELSE
DBMS_OUTPUT.PUT_LINE('EMPLOYEEDIDNOTGETANYCOMMISSION');
END    IF;
DBMS_OUTPUT.PUT_LINE(EMP_CUR%ROWCOUNT||'RECORDSFETCHEDSOFAR');     CLOSE
EMP_CUR;
END;
/
```

OUTPUT

```
EMPLOYEE: MARTIN GETS COMMISSION: 1400
1 RECORDS FETCHED SO FAR

PL/SQL procedure successfully completed.
```

6. PL/SQLTOPMPLEMENT%FOUNDUSINGDEPTTABLE.

```
SQL>SETSERVEROUTPUTON;

DECLARE

CURSORDEPTCURISSELECTDEPTNO,DNAMEFROMDEPTWHERE
DEPTNO    =    10;

TDEPTCURDEPTCUR%ROWTYPE;

BEGIN

OPENDEPTCUR;

FETCH    DEPTCUR    INTO

TDEPTCUR;    IF

DEPTCUR%FOUNDTHEN

DBMS_OUTPUT.PUT_LINE('DEPTNO:'||TDEPTCUR.DEPTNO||'DNAME:'||
TDEPTCUR.DNAME);

ELSE

DBMS_OUTPUT.PUT_LINE('No    department    found

withDEPTNO    10');    END    IF;

CLOSEDEPTCUR;

END;

/
```

OUTPUT

```
DEPTNO: 10  DNAME: ACCOUNTING

PL/SQL procedure successfully completed.
```

-

7. PL/SQL BLOCK USINGTHE COURSOR TO DISPLAY DETAILS OF ALL EMPLOYEE'SFROMEMPTABLEWHOSESUMOFSALANDCOMMIS >2000

```
SQL>SETSERVEROUTPUTON;
DECLARE
MEM_PNO     EMP.EMPNO%TYPE;
MEM_NAMEEMP.ENAME%TYPE;
MEM_SAL    EMP.SAL%TYPE;
CURSOR    C1    IS
SELECTEMPNO,ENAME,SAL+NVL(COMM,0)
FROM    EMP
WHERESAL+NVL(COMM,0)>2000;
 BEGIN
OPENC1;
DBMS_OUTPUT.PUT_LINE('EMPNO    ENAME    MSAL');
LOOP
FETCHC1INTOMEM_PNO,MEM_NAME,MEM_SAL;
EXIT    WHEN    C1%NOTFOUND;
DBMS_OUTPUT.PUT_LINE(MEM_PNO||''||MEM_NAME     ||''||MEM_SAL);
END    LOOP;
CLOSEC1;
END;
/
```

<u>OUTPUT</u>

```
EMPNO  ENAME   MSAL
7839  KING  5000
7698  BLAKE  2850
7782  CLARK  2450
7566  JONES  2975
7902  FORD  3000
7654  MARTIN  2650

PL/SQL procedure successfully completed.
```

8. **PL/SQL BLOCK THE RETRIVES COURSE TABLE INTO CURSOR.INPUT CODEIDTOSEARCH.IFCOURSEEXISTPRINTTHEINFORMATION.IF COURSE ID DOES NOT EXIST, PRINT USER DEFINED MESSAGE BY THROWING USER DEFINED EXCEPTION.**

```
SQL>SETSERVEROUTPUTON;
DECLARE
  C_NAMECOURSE.COURSENAME%TYPE;
COURSORC1ISSELECTCOURSENAMEFROMCOURSEWHERE
COURSEIS:='&ID';
COURSE_NOT_FOUND
EXCEPTION;
BEGIN
OPENC1;
FETCHC1INTOC_NAME;
IF C1% FOUND THEN
DBMS_OUTPUT.PUT_LINE('COURSE NAME
IS'||C_NAME);
ELSERAISECOURSE_NOT_FOUND;
END IF;
EXCEPTIONWHENCOURSE_NOT_FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('COURSE ID NOT
FOUND');
END;
/
```

<u>OUTPUT</u>

```
Enter value for id: 13
old 3: cursorcl is select coursename from course where courseis=&id'
new 3:cursor cl is select coursename from course where courseis='13';
course name isB.Sc.computeres
PL/SQL procedure successfully completed.
SQL>/
Enter value for id: 18
old 3: cursor cl is select coursename from course where courseis=&id';
new 3:cursor c1 is select coursename from course where courseis=18';
course id is not found
PL/SQL procedure successfully completed
```

9. PL/SQLPROGRAMTOHANDLEZERODIVIDEEXCEPTION

```
SQL>SETSERVEROUTPUTON;
DECLARE
xNUMBER:=&X;
Y       NUMBER
:=    &Y;    BEGIN
IF    Y    <>    0    THEN
DBMS_OUTPUT.PUT_LINE('DIVISION='||X/
Y);    ELSE
DBMS_OUTPUT.PUT_LINE('CHECK     THE
DENOMINATOR');    END    IF;
END;
 /
```

OUTPUT:

```
Enter value for x: 23
old    2:    X NUMBER := &X;
new    2:    X NUMBER := 23;
Enter value for y: 9
old    3:    Y NUMBER := &Y;
new    3:    Y NUMBER := 9;
DIVISION=2.5555555555555555555555555555555555556

PL/SQL procedure successfully completed.
```

10. PL/SQLTHATCREATESATRIGGERTHATINSERTSORUPDATES THE VALUES OF ENAME.JOB AS LOWER-CASE STRING.

```
SQL> CREATE OR REPLACE TRIGGER
LOWERNAMEBEFOREINSERTORUPDATEONEMP
FOR EACH ROW BEGIN
:NEW.ENAME:=:LOWER(“NEW.ENAME);
NEW.JOB:=LOWER(:MEW.JOB);
END;
/
```

OUTPUT:

Triggercreated.

SQL>UPDATEEEMPSETJOB=’MANAGER’WHEREENAME=SMITH;

```
SQL>  UPDATE EMP SET JOB=’MANAGER’ WHERE ENAME='SMITH';

1 row updated.
```

11. PL/SQLFORSIMPLELOOP.

```
SQL>SETSERVEROUTPUTON;
DECLARE
INUMBER:=0;
BEGIN
LOOP
DBMS_OUTPUT.PUT_LINE('I='||I);
I:=I+2;
EXITWHENI>10;
END LOOP;
END;
/
```

OUTPUT:

```
I=0
I=2
I=4
I=6
I=8
I=10

PL/SQL procedure successfully completed.
```

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During 3rd year 1st semester of the academic year 2025-2026

Name: Macha Divya

UIDNo:111723016011

Class: DCSIT

Signature of Internal **Signature of HOD**

Signature of External **Signature of Principal**

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STRUCTUREDQUERYLANGUAGE

1) **WRITE A QUERY TO CREATE A TABLE WITH THE FOLLOWINGPARAMETERSSNO,SNAME,DOB,SUB1,SUB2,AND SUB3 USING A APPROPRIATE DATA TYPE.**

SQL>CREATETABLESTUD1(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>DESC

STUD1

```
Table created.

SQL> desc stud1
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                                NUMBER(11)
 SNAME                                     NOT NULL VARCHAR2(13)
 DOB                                                DATE
 SUB1                                               CHAR(8)
 SUB2                                               CHAR(8)
 SUB3                                               CHAR(8)
```

2) WRITEAQUERYTOINSERTDATAINTOTABLEANDDISPLAYTHETABLE.

SQL>INSERTINTOSTUD1VALUES(&SNO,’&SNAME’,’&DOB’,’&SUB1’,’2&SUB2’,’&SUB3’);

```
SQL> insert into stud1 values(&sno,’&sname’,’&dob’,’&sub1’,’2 &sub2’,’&sub3’);
Enter value for sno: 1
Enter value for sname: sailaja
Enter value for dob: 3-oct-1994
Enter value for sub1: Eng
Enter value for sub2: Litera
Enter value for sub3: Phylo
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(1,'sailaja','3-oct-1994','Eng','2 Litera','Phylo')

1 row created.

SQL> /
Enter value for sno: 2
Enter value for sname: shanmukh
Enter value for dob: 11-mar-1999
Enter value for sub1: phy
Enter value for sub2: ldc
Enter value for sub3: m1
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(2,'shanmukh','11-mar-1999','phy','2 ldc','m1')

1 row created.
SQL> /
Enter value for sno: 3
Enter value for sname: hardik jain
Enter value for dob: 02-oct-2005
Enter value for sub1: engg
Enter value for sub2: m2
Enter value for sub3: pis
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(3,'hardik jain','02-oct-2005','engg','2 m2','pis')

1 row created.
```

SQL>SELECT*FROMSTUD1;

```
SQL> select * from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

3) **WRITEAQUERYTODISPLAYTHEDATAINTABLEASI.SELECTROWSAND COLUMNS II. SELECTED COLUMNS AND ALL ROWS ALL ROWS III. ALL ROWS AND ALL COLUMNS IV. SELECTED ROWS AND ALL COLUMNS**

I. SQL>SELECT*FROMSTUD1WHERESNO>2;

```
SQL> select * from stud1 where sno>2;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

II. SQL>SELECTSNO,SNAMEFROMSTUD1;

```
SQL> select sno,sname from stud1;

       SNO SNAME
---------- -------------
         1 sailaja
         2 shanmukh
         3 hardik jain
```

III. SQL>SELECT*FROMSTUD1;

```
SQL> select *from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

IV. SQL>SELECTSNO,SNAMEFROMSTUD1WHERESNO=1;

```
SQL> select sno,sname from stud1 where sno=1;

       SNO SNAME
---------- -------------
         1 sailaja
```

4) **WRITEAQUERYTOELIMINATEDUPLICATEROWSUSINGSELECT STATEMENT**.

SQL>SELECTDISTINCTSUB1FROMSTUD1;

```
SQL> select distinct sub1 from stud1;

SUB1
--------
engg
Eng
phy
```

5) **WRITEAQUERYTOSORTTHEDATAIN A TABLE**.

SQL>SELECT*FROMSTUD1ORDERBYSNO DESC;

```
SQL> select *from stud1 order by sno desc;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
```

6) **WRITEAQUERYTOCREATETABLEFROM**

ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.

SQL>SELECT*FROMSTUD

```
SQL> select * from stud4;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

7) WRITEAQUERYTOCREATETABLEFROM ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.
SQL>SELECT*FROMSTUD4;

8) WRITEAQUERYTOINSERTDATAINTOATABLEFROMANOTHERTABLE

SQL>CREATETABLESTUD6(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>INSERTINTOSTUD6SELECT*FROMSTUD1WHERE

SNO<=2; 2 ROWS CREATED.

SQL>SELECT*FROMSTUD6;

```
SQL> select * from stud6;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
```

9) WRITEAQUERYTODELETEALLROWS FROM A TABLE.

SQL>DELETEFROMSTUD4;

```
SQL> delete from stud4;

4 rows deleted.
```

10) WRITEAQUERYTOMODIFYTHEEXISTING COLUMNS IN A TABLE.

SQL>SELECTSNAMEASNAMEFROMSTUD1;

```
SQL> select sname as name from stud1;

NAME
--------------------
sailaja
B Samaresh
BS
```

11) WRITEAQUERYTO TRUNCATE A TABLE.

SQL>TRUNCATETABLESTUD1;

TABLE TRUNCATED.

12) WRITEAQUERYTO DESTROY A TABLE.

SQL>DROPTABLESTUD1;

TABLE
DROPPED.

13) WRITEAQUERYTOCREATEATABLECUSTOMERWITHCUSTIDAS PRIMARY KEY FOLLOWED BY OTHER COLUMNS.

SQL>CREATETABLECUSTOMER(CUSTIDNUMBER(4)CONSTRAINTPKLPRIMARY KEY,PRODUCT VARCHAR2(5));

TABLECREATED.

14) WRITEAQUERYTOESTABLISHFOREIGNKEYCONSTRAINT.

SQL> CREATE TABLE PRODUCER (PRODUCT_ID NUMBER (8) UNIQUE,CUST_DEMANDNUMBER(4)REFERENCESCUSTOMER(CUSTID));

TABLECREATED.

15) WRITEAQUERYTOCALCULATETHEANNUALSALARYOFAN EMPLOYEE

SQL>SELECTEMPNO,SAL*12FROMEMP;

```
SQL> select empno,sal*12 from emp;

     EMPNO     SAL*12
---------- ----------
      7839      60000
      7698      34200
      7782      29400
      7566      35700
      7788      36000
      7902      36000
      7369       9600
      7499      19200
      7521      15000
      7654      15000
      7844      18000

     EMPNO     SAL*12
---------- ----------
      7876      13200
      7900      11400
      7934      15600

14 rows selected.
```

16) WRITEAQUERYTOCREATEATABLEHAVINGBNAMEAND BNO, AND DISPLAY VALUE WHERE BNO IS NULL.

SQL>CREATETABLEDEMO1(BNONUMBER(10),BNAME

CHAR(20)); TABLE CREATED.

SQL> INSERT INTO DEMO1
VALUES(1,'MADHU');1ROWCREATED.

SQL> INSERT INTO DEMO1
VALUES(NULL,'SINDHU');1 ROWCREATED.

SQL>INSERTINTODEMO1
VALUES(0,'HEMA');
1ROWCREATED.

SQL>SELECT*FROMDEMO1WHEREBNOISNULL;

```
SQL> select * from demo1 where bno is null;

       BNO BNAME
---------- --------------------
           sindhu
```

17) WRITEAQUERYTOCREATEATABLETHATDOESNOTTAKENULL VALUE.

SQL>CREATETABLEDEMO2(NONUMBER(10),NAMECHAR(20)NOTNULL); TABLE CREATED.

SQL>INSERTINTODEMO2VALUES(1,NULL);

```
SQL> insert into demo2 values(1,null);
insert into demo2 values(1,null)
                          *
ERROR at line 1:
ORA-01400: cannot insert NULL into ("SYSTEM"."DEMO2"."NAME")
```

18) WRITEAQUERYTOIMPLEMENTCONSTRAINTS.

SQL>CREATETABLESTU(SNONUMBER(10)PRIMARYKEY,SNAME CHAR(30) UNIQUE,PHNO NUMBER(10)CHECK(PHNO>800000000),ADDRESSCHAR(10)NOTNULL);

TABLE CREATED.

SQL>DESCSTU;

```
SQL> desc stu;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                       NOT NULL NUMBER(10)
 SNAME                                              CHAR(30)
 PHNO                                               NUMBER(10)
 ADDRESS                                   NOT NULL CHAR(10)
```

19) WRITEAQUERYTODROPPRIMARYKEY CONSTRAINT FROM A TABLE.

SQL>ALTERTABLESTUDROPPRIMARY KEY; TABLE ALTERED.

20) WRITEAQUERYTOIMPLEMENTOR.

SQL>SELECT*FROMEMPWHEREENAME=’FORD’ORSAL=3000;

```
SQL> SELECT * FROM EMP WHERE eNAME='FORD' OR SAL=3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

20) WRITEAQUERYTOIMPLEMENTNOT.

SQL>SELECT*FROMEMPWHERENO SAL>2400;

```
SQL> select * from emp where not SAL>2400;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7369 SMITH      CLERK           7902 17-DEC-80        800
        20

      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300
        30

      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500
        30


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400
        30

      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0
        30

      7876 ADAMS      CLERK           7788 23-MAY-87       1100
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7900 JAMES      CLERK           7698 03-DEC-81        950
        30

      7934 MILLER     CLERK           7782 23-JAN-82       1300
        10
```

21) WRITEAQUERYTOIMPLEMENTRANGE SEARCHING.

SQL>SELECT*FROMEMPWHERESALBETWEEN2000 AND 3000;

```
SQL> select * from emp where sal between 2000 and 3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7782 CLARK      MANAGER         7839 09-JUN-81       2450
        10

      7566 JONES      MANAGER         7839 02-APR-81       2975
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7788 SCOTT      ANALYST         7566 19-APR-87       3000
        20

      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

22) WRITEAQUERYTODISPLAYTHEEMPLOYEENAMETHAT BEGINS WITH Q.

SQL>SELECT*FROMEMPWHEREENAMELIKE'%Q'; NO ROWS SELECTED.

23) WRITEAQUERYTOIMPLEMENTSYSTEM DATE.

SQL>SELECTSYSDATEFROMDUEAL;

```
SQL> SELECT SYSDATE FROM DUAL

SYSDATE
---------
02-OCT-24
```

24) WRITEAQUERYTOFINDTHEAVERAGESALARY OF EMP TABLE

SQL>SELECTAVG(SAL)“AVERAGESALARY”FROMEMP;

```
SQL> SELECT AVG(SAL) “AVERAGE SALARY” FROM EMP;

AVERAGE SALARY
--------------
    2073.21429
```

25) WRITEAQUERYTOFINDTHEMINIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMIN(SAL)“MINIMUMSALARY”FROMEMP;

```
SQL> SELECT MIN(SAL) “MINIMUM SALARY” FROM EMP;

MINIMUM SALARY
--------------
           800
```

26) WRITEAQUERYTOFINDTHEMAXIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMAX(SAL)“MAXIMUMSALARY”FROMEMP;

```
SQL> SELECT MAX(SAL) “MAXIMUM SALARY” FROM EMP;

MAXIMUM SALARY
--------------
          5000
```

27) WRITEAQUERYTOIMPLEMENTCOUNT FUNCTION ON EMP TABLE.

SQL>SELECTCOUNT(*)“TOTALNOOFROWS”FROMEMP;

```
SQL> SELECT COUNT(*) “TOTAL NO OF ROWS” FROM EMP;

TOTAL NO OF ROWS
----------------
              14
```

28) WRITEAQUERYTOIMPLEMENTSUM FUNCTION ON EMP TABLE.

SQL>SELECTSUM(SAL)”SUMOFSLARY”FROMEMP;

```
SQL> SELECT SUM(SAL)”SUM OF SLARY” FROM EMP;

SUM OF SLARY
------------
       29025
```

PL/SQLQUERIES

1. ADD2NUMBERS.

```
SQL>SETSERVEROUTPUTON;
DECLARE
VAR1INTEGER;
VAR2INTEGER;
VAR3INTEGER;
BEGIN
VAR1    :=
&VAR1;VAR2    :=
&VAR2;VAR3:=VAR1+
VAR2;
DBMS_OUTPUT.PUT_LINE('THESUM     OF'||VAR1    ||'AND    '||VAR2    ||    'IS:'||VAR3);
END;
/
```

OUTPUT:

```
Enter value for var1: 10
old   6:   VAR1 := &VAR1;
new   6:   VAR1 := 10;
Enter value for var2: 20
old   7:   VAR2 := &VAR2;
new   7:   VAR2 := 20;
The sum of 10 and 20 is: 30

PL/SQL procedure successfully completed.
```

2. FORLOOPDEMO

```
SQL>SETSERVEROUTPUT
ON;    DECALARE
    INUMBER(2);
    BEGIN
    FORIIN1..10
    LOOP
    DBMS_OUTPUT.PUT_LINE(
    I);    END    LOOP;
    END;
```

OUTPUT:

```
1
2
3
4
5
6
7
8
9
10

PL/SQL procedure successfully completed.
```

3. PL/SQLPROGRAMTOSWAPTWONUMBERS

i. USINGTEMP

```
SQL>SETSERVEROUTPUTON;
DECLARE
 ANUMBER(2):=&A;
BNUMBER(2):=&B;
TEMP
NUMBER(2);
BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE    SWAPPING
 A='||A||'B='||B);    TEMP:=A;
B:=TEMP;
DBMS_OUTPUT.PUT_LINE('AFTER    SWAPPING
 A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 10
old   2:   A NUMBER(2):=&A;
new   2:   A NUMBER(2):=10;
Enter value for b: 20
old   3:   B NUMBER(2):=&B;
new   3:   B NUMBER(2):=20;
BEFORE SWAPPING A=10B=20
AFTER SWAPPING A=10B=10

PL/SQL procedure successfully completed.
```

ii. WITHOUTUSINGTEMP

```
SQL>SETSERVEROUTPUTON;

 DECLARE
1.  NUMBER(2):=&A;
2.  NUMBER
 (2):=&B;    BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE
SWAPPING’A='||A||'B='||B);
A:=A+B;    B:=A-B;
A:=A-B;
DBMS_OUTPUT.PUT_LINE('AFTER
SWAPPING’A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 4
old   2:   A NUMBER(2) := &A;
new   2:   A NUMBER(2) := 4;
Enter value for b: 5
old   3:   B NUMBER(2) := &B;
new   3:   B NUMBER(2) := 5;
BEFORE SWAPPING A=4 B=5
AFTER SWAPPING A=5 B=4

PL/SQL procedure successfully completed.
```

4. WRITEAPROGRAMTOPRINTEMPNO,ENAME,JOB,SALARYOFAN EMPLOYEE OF GIVEN EMPNO.

```
SQL>SETSERVEROUTPUTON;
DECLARE
TYPE EMPREC IS RECORD (ENO
EMP.EMPNO%TYPE,NEMP.ENAME%TYPE,J
EMP.JOB%TYPE,S EMP.SAL%TYPE);
R EMPREC;
ENOEMP.EMPNO%TYPE:=&ENO;
BEGIN
SELECTEMPNO,ENAME,JOB,SAL
INTO R
FROMEMP
WHERE EMPNO = ENO; -- Corrected the
condition
DBMS_OUTPUT.PUT_LINE(R.ENO||''||R.N||''||R.J
||''||R.S);
END;
/
```

OUTPUT:

```
Enter value for eno: 7839
old   4: ENO EMP.EMPNO%TYPE := &ENO;
new   4: ENO EMP.EMPNO%TYPE := 7839;
7839 KING PRESIDENT 5000

PL/SQL procedure successfully completed.
```

5. PL/SQLPROGRAMTOIMPLEMENT%FOUND

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSOREMP_CUR
ISSELECTENAME,COMM
FROMEMPWHEREEMPNO=7654
AND    COMM    IS    NOT    NULL;
EMP_NAME    VARCHAR2(20);
EMP_COMM    NUMBER(10,2);
BEGIN    OPEN    EMP_CUR;
FETCHEMP_CURINTOEMP_NAME,EMP_COMM;
IF    EMP_CUR%FOUND    THEN
DBMS_OUTPUT.PUT_LINE('EMPLOYEE:'||EMP_NAME     ||'GETSCOMMISSION:'||
EMP_COMM);
ELSE
DBMS_OUTPUT.PUT_LINE('EMPLOYEEDIDNOTGETANYCOMMISSION');
END    IF;
DBMS_OUTPUT.PUT_LINE(EMP_CUR%ROWCOUNT||'RECORDSFETCHEDSOFAR');     CLOSE
EMP_CUR;
END;
/
```

OUTPUT

```
EMPLOYEE: MARTIN GETS COMMISSION: 1400
1 RECORDS FETCHED SO FAR

PL/SQL procedure successfully completed.
```

6. PL/SQLTOPMPLEMENT%FOUNDUSINGDEPTTABLE.

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSORDEPTCURISSELECTDEPTNO,DNAMEFROMDEPTWHERE
DEPTNO = 10;
TDEPTCURDEPTCUR%ROWTYPE;
BEGIN
OPENDEPTCUR;
FETCH DEPTCUR INTO
TDEPTCUR; IF
DEPTCUR%FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('DEPTNO:'||TDEPTCUR.DEPTNO||'DNAME:'||
TDEPTCUR.DNAME);
ELSE
DBMS_OUTPUT.PUT_LINE('No department found
withDEPTNO 10'); END IF;
CLOSEDEPTCUR;
END;
/
```

OUTPUT

```
DEPTNO: 10  DNAME: ACCOUNTING

PL/SQL procedure successfully completed.
```

7. PL/SQL BLOCK USINGTHE COURSOR TO DISPLAY DETAILS OF ALL EMPLOYEE'SFROMEMPTABLEWHOSESUMOFSALANDCOMMIS >2000

```
SQL>SETSERVEROUTPUTON;
DECLARE
MEM_PNO    EMP.EMPNO%TYPE;
MEM_NAMEEMP.ENAME%TYPE;
MEM_SAL    EMP.SAL%TYPE;
CURSOR    C1    IS
SELECTEMPNO,ENAME,SAL+NVL(COMM,0)
FROM    EMP
WHERESAL+NVL(COMM,0)>2000;
 BEGIN
OPENC1;
DBMS_OUTPUT.PUT_LINE('EMPNO    ENAME    MSAL');
LOOP
FETCHC1INTOMEM_PNO,MEM_NAME,MEM_SAL;
EXIT    WHEN    C1%NOTFOUND;
DBMS_OUTPUT.PUT_LINE(MEM_PNO||"||MEM_NAME     ||"||MEM_SAL);
END    LOOP;
CLOSEC1;
END;
/
```

<u>OUTPUT</u>

```
EMPNO  ENAME   MSAL
7839  KING  5000
7698  BLAKE  2850
7782  CLARK  2450
7566  JONES  2975
7902  FORD  3000
7654  MARTIN  2650

PL/SQL procedure successfully completed.
```

8. **PL/SQL BLOCK THE RETRIVES COURSE TABLE INTO CURSOR.INPUT CODEIDTOSEARCH.IFCOURSEEXISTPRINTTHEINFORMATION.IF COURSE ID DOES NOT EXIST, PRINT USER DEFINED MESSAGE BY THROWING USER DEFINED EXCEPTION.**

```
SQL>SETSERVEROUTPUTON;
DECLARE
  C_NAMECOURSE.COURSENAME%TYPE;
COURSORC1ISSELECTCOURSENAMEFROMCOURSEWHERE
COURSEIS:='&ID';
COURSE_NOT_FOUND
EXCEPTION;

BEGIN
OPENC1;
FETCHC1INTOC_NAME;
IF  C1%  FOUND  THEN
DBMS_OUTPUT.PUT_LINE('COURSE  NAME
IS'||C_NAME);
ELSERAISECOURSE_NOT_FOUND;
END  IF;
EXCEPTIONWHENCOURSE_NOT_FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('COURSE  ID  NOT
FOUND');
END;
/
```

<u>OUTPUT</u>

```
Enter value for id: 13
old 3: cursorcl is select coursename from course where courseis=&id'
new 3:cursor cl is select coursename from course where courseis='13';
course name isB.Sc.computeres
PL/SQL procedure successfully completed.
SQL>/
Enter value for id: 18
old 3: cursor cl is select coursename from course where courseis=&id';
new 3:cursor c1 is select coursename from course where courseis=18';
course id is not found
PL/SQL procedure successfully completed
```

9. PL/SQLPROGRAMTOHANDLEZERODIVIDEEXCEPTION

```
SQL>SETSERVEROUTPUTON;
DECLARE
xNUMBER:=&X;
Y       NUMBER
:=   &Y;   BEGIN
IF   Y   <>   0   THEN
DBMS_OUTPUT.PUT_LINE('DIVISION='||X/
Y);   ELSE
DBMS_OUTPUT.PUT_LINE('CHECK    THE
DENOMINATOR');   END   IF;
END;
 /
```

OUTPUT:

```
Enter value for x: 23
old   2:   X NUMBER := &X;
new   2:   X NUMBER := 23;
Enter value for y: 9
old   3:   Y NUMBER := &Y;
new   3:   Y NUMBER := 9;
DIVISION=2.5555555555555555555555555555555555556

PL/SQL procedure successfully completed.
```

10. PL/SQLTHATCREATESATRIGGERTHATINSERTSORUPDATES THE VALUES OF ENAME.JOB AS LOWER-CASE STRING.

```
SQL> CREATE OR REPLACE TRIGGER
LOWERNAMEBEFOREINSERTORUPDATEONEMP
FOR EACH ROW BEGIN
:NEW.ENAME:=:LOWER(“NEW.ENAME);
NEW.JOB:=LOWER(:MEW.JOB);
END;
/
```

OUTPUT:

Triggercreated.

SQL>UPDATEEEMPSETJOB=’MANAGER’WHEREENAME=SMITH;

```
SQL>  UPDATE EMP SET JOB=’MANAGER’ WHERE ENAME='SMITH';

1 row updated.
```

11. PL/SQLFORSIMPLELOOP.

```
SQL>SETSERVEROUTPUTON;
DECLARE
INUMBER:=0;
BEGIN
LOOP
DBMS_OUTPUT.PUT_LINE('I='||I);
I:=I+2;
EXITWHENI>10;
END LOOP;
END;
/
```

OUTPUT:

```
I=0
I=2
I=4
I=6
I=8
I=10

PL/SQL procedure successfully completed.
```

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During 3rd year 1st semester of the academic year 2025-2026

Name: Datla Srihitha

UIDNo:111723016005

Class:DCSIT

Signature of Internal **Signature of HOD**

Signature of External **Signature of Principal**

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STRUCTUREDQUERYLANGUAGE

1) **WRITE A QUERY TO CREATE A TABLE WITH THE FOLLOWINGPARAMETERSSNO,SNAME,DOB,SUB1,SUB2,AND SUB3 USING A APPROPRIATE DATA TYPE.**

SQL>CREATETABLESTUD1(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>DESC

STUD1

```
Table created.

SQL> desc stud1
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                                NUMBER(11)
 SNAME                                     NOT NULL VARCHAR2(13)
 DOB                                                DATE
 SUB1                                               CHAR(8)
 SUB2                                               CHAR(8)
 SUB3                                               CHAR(8)
```

2) WRITEAQUERYTOINSERTDATAINTOTABLEANDDISPLAYTHETABLE.

SQL>INSERTINTOSTUD1VALUES(&SNO,’&SNAME’,’&DOB’,’&SUB1’,’2&SUB2’,’&SUB3’);

```
SQL> insert into stud1 values(&sno,’&sname’,’&dob’,’&sub1’,’2 &sub2’,’&sub3’);
Enter value for sno: 1
Enter value for sname: sailaja
Enter value for dob: 3-oct-1994
Enter value for sub1: Eng
Enter value for sub2: Litera
Enter value for sub3: Phylo
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(1,'sailaja','3-oct-1994','Eng','2 Litera','Phylo')

1 row created.

SQL> /
Enter value for sno: 2
Enter value for sname: shanmukh
Enter value for dob: 11-mar-1999
Enter value for sub1: phy
Enter value for sub2: ldc
Enter value for sub3: m1
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(2,'shanmukh','11-mar-1999','phy','2 ldc','m1')

1 row created.
SQL> /
Enter value for sno: 3
Enter value for sname: hardik jain
Enter value for dob: 02-oct-2005
Enter value for sub1: engg
Enter value for sub2: m2
Enter value for sub3: pis
old   1: insert into stud1 values(&sno,'&sname','&dob','&sub1','2 &sub2','&sub3')
new   1: insert into stud1 values(3,'hardik jain','02-oct-2005','engg','2 m2','pis')

1 row created.
```

SQL>SELECT*FROMSTUD1;

```
SQL> select * from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

3) **WRITEAQUERYTODISPLAYTHEDATAINTABLEASI.SELECTROWSAND COLUMNS II. SELECTED COLUMNS AND ALL ROWS ALL ROWS III. ALL ROWS AND ALL COLUMNS IV. SELECTED ROWS AND ALL COLUMNS**

I. SQL>SELECT*FROMSTUD1WHERESNO>2;

```
SQL> select * from stud1 where sno>2;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

II. SQL>SELECTSNO,SNAMEFROMSTUD1;

```
SQL> select sno,sname from stud1;

       SNO SNAME
---------- -------------
         1 sailaja
         2 shanmukh
         3 hardik jain
```

III. SQL>SELECT*FROMSTUD1;

```
SQL> select *from stud1;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

IV. SQL>SELECTSNO,SNAMEFROMSTUD1WHERESNO=1;

```
SQL> select sno,sname from stud1 where sno=1;

       SNO SNAME
---------- -------------
         1 sailaja
```

4) **WRITEAQUERYTOELIMINATEDUPLICATEROWSUSINGSELECT STATEMENT**.

SQL>SELECTDISTINCTSUB1FROMSTUD1;

```
SQL> select distinct sub1 from stud1;

SUB1
--------
engg
Eng
phy
```

5) **WRITEAQUERYTOSORTTHEDATAIN A TABLE**.

SQL>SELECT*FROMSTUD1ORDERBYSNO DESC;

```
SQL> select *from stud1 order by sno desc;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         3 hardik jain   02-OCT-05 engg     2 m2     pis
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
```

6) **WRITEAQUERYTOCREATETABLEFROM**

ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.

SQL>SELECT*FROMSTUD

```
SQL> select * from stud4;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
         3 hardik jain   02-OCT-05 engg     2 m2     pis
```

7) WRITEAQUERYTOCREATETABLEFROM ANOTHER TABLE.

SQL>CREATE TABLESTUD4ASSELECT*FROMSTUD1;

TABLECREATED.
SQL>SELECT*FROMSTUD4;

8) WRITEAQUERYTOINSERTDATAINTOATABLEFROMANOTHERTABLE

SQL>CREATETABLESTUD6(SNONUMBER(11)UNIQUE,SNAMEVARCHAR(13)NOT NULL,DOB DATE,SUB1 CHAR(8),SUB2 CHAR(8),SUB3 CHAR(8));

TABLECREATED.

SQL>INSERTINTOSTUD6SELECT*FROMSTUD1WHERE

SNO<=2; 2 ROWS CREATED.

SQL>SELECT*FROMSTUD6;

```
SQL> select * from stud6;

       SNO SNAME         DOB       SUB1     SUB2     SUB3
---------- ------------- --------- -------- -------- --------
         1 sailaja       03-OCT-94 Eng      2 Litera Phylo
         2 shanmukh      11-MAR-99 phy      2 ldc    m1
```

9) WRITEAQUERYTODELETEALLROWS FROM A TABLE.

SQL>DELETEFROMSTUD4;

```
SQL> delete from stud4;

4 rows deleted.
```

10) WRITEAQUERYTOMODIFYTHEEXISTING COLUMNS IN A TABLE.

SQL>SELECTSNAMEASNAMEFROMSTUD1;

```
SQL> select sname as name from stud1;

NAME
--------------------
sailaja
B Samaresh
BS
```

11) WRITEAQUERYTO TRUNCATE A TABLE.

SQL>TRUNCATETABLESTUD1;

TABLE TRUNCATED.

12) WRITEAQUERYTO DESTROY A TABLE.

SQL>DROPTABLESTUD1;

TABLE
DROPPED.

13) WRITEAQUERYTOCREATEATABLECUSTOMERWITHCUSTIDAS PRIMARY KEY FOLLOWED BY OTHER COLUMNS.

SQL>CREATETABLECUSTOMER(CUSTIDNUMBER(4)CONSTRAINTPKLPRIMARY KEY,PRODUCT VARCHAR2(5));

TABLECREATED.

14) WRITEAQUERYTOESTABLISHFOREIGNKEYCONSTRAINT.

SQL> CREATE TABLE PRODUCER (PRODUCT_ID NUMBER (8) UNIQUE,CUST_DEMANDNUMBER(4)REFERENCESCUSTOMER(CUSTID));

TABLECREATED.

15) WRITEAQUERYTOCALCULATETHEANNUALSALARYOFAN EMPLOYEE

SQL>SELECTEMPNO,SAL*12FROMEMP;

```
SQL> select empno,sal*12 from emp;

     EMPNO     SAL*12
---------- ----------
      7839      60000
      7698      34200
      7782      29400
      7566      35700
      7788      36000
      7902      36000
      7369       9600
      7499      19200
      7521      15000
      7654      15000
      7844      18000

     EMPNO     SAL*12
---------- ----------
      7876      13200
      7900      11400
      7934      15600

14 rows selected.
```

16) WRITEAQUERYTOCREATEATABLEHAVINGBNAMEAND BNO, AND DISPLAY VALUE WHERE BNO IS NULL.

SQL>CREATETABLEDEMO1(BNONUMBER(10),BNAME

CHAR(20)); TABLE CREATED.

SQL> INSERT INTO DEMO1
VALUES(1,'MADHU');1ROWCREATED.

SQL> INSERT INTO DEMO1
VALUES(NULL,'SINDHU');1 ROWCREATED.

SQL>INSERTINTODEMO1
VALUES(0,'HEMA');
1ROWCREATED.

SQL>SELECT*FROMDEMO1WHEREBNOISNULL;

```
SQL> select * from demo1 where bno is null;

       BNO BNAME
---------- --------------------
           sindhu
```

17) WRITEAQUERYTOCREATEATABLETHATDOESNOTTAKENULL VALUE.

SQL>CREATETABLEDEMO2(NONUMBER(10),NAMECHAR(20)NOTNULL); TABLE CREATED.

SQL>INSERTINTODEMO2VALUES(1,NULL);

```
SQL> insert into demo2 values(1,null);
insert into demo2 values(1,null)
                          *
ERROR at line 1:
ORA-01400: cannot insert NULL into ("SYSTEM"."DEMO2"."NAME")
```

18) WRITEAQUERYTOIMPLEMENTCONSTRAINTS.

SQL>CREATETABLESTU(SNONUMBER(10)PRIMARYKEY,SNAME CHAR(30) UNIQUE,PHNO NUMBER(10)CHECK(PHNO>800000000),ADDRESSCHAR(10)NOTNULL);

TABLE CREATED.

SQL>DESCSTU;

```
SQL> desc stu;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                       NOT NULL NUMBER(10)
 SNAME                                              CHAR(30)
 PHNO                                               NUMBER(10)
 ADDRESS                                   NOT NULL CHAR(10)
```

19) WRITEAQUERYTODROPPRIMARYKEY CONSTRAINT FROM A TABLE.

SQL>ALTERTABLESTUDROPPRIMARY KEY; TABLE ALTERED.

20) WRITEAQUERYTOIMPLEMENTOR.

SQL>SELECT*FROMEMPWHEREENAME=’FORD’ORSAL=3000;

```
SQL> SELECT * FROM EMP WHERE eNAME='FORD' OR SAL=3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

20) WRITEAQUERYTOIMPLEMENTNOT.

SQL>SELECT*FROMEMPWHERENO SAL>2400;

```
SQL> select * from emp where not SAL>2400;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7369 SMITH      CLERK           7902 17-DEC-80        800
        20

      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300
        30

      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500
        30


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400
        30

      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0
        30

      7876 ADAMS      CLERK           7788 23-MAY-87       1100
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7900 JAMES      CLERK           7698 03-DEC-81        950
        30

      7934 MILLER     CLERK           7782 23-JAN-82       1300
        10
```

21) WRITEAQUERYTOIMPLEMENTRANGE SEARCHING.

SQL>SELECT*FROMEMPWHERESALBETWEEN2000 AND 3000;

```
SQL> select * from emp where sal between 2000 and 3000;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7782 CLARK      MANAGER         7839 09-JUN-81       2450
        10

      7566 JONES      MANAGER         7839 02-APR-81       2975
        20


     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTNO
----------
      7788 SCOTT      ANALYST         7566 19-APR-87       3000
        20

      7902 FORD       ANALYST         7566 03-DEC-81       3000
        20
```

22) WRITEAQUERYTODISPLAYTHEEMPLOYEENAMETHAT BEGINS WITH Q.

SQL>SELECT*FROMEMPWHEREENAMELIKE'%Q'; NO ROWS SELECTED.

23) WRITEAQUERYTOIMPLEMENTSYSTEM DATE.

SQL>SELECTSYSDATEFROMDUEAL;

```
SQL> SELECT SYSDATE FROM DUAL

SYSDATE
---------
02-OCT-24
```

24) WRITEAQUERYTOFINDTHEAVERAGESALARY OF EMP TABLE

SQL>SELECTAVG(SAL)“AVERAGESALARY”FROMEMP;

```
SQL> SELECT AVG(SAL) “AVERAGE SALARY” FROM EMP;

AVERAGE SALARY
--------------
    2073.21429
```

25) WRITEAQUERYTOFINDTHEMINIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMIN(SAL)“MINIMUMSALARY”FROMEMP;

```
SQL> SELECT MIN(SAL) “MINIMUM SALARY” FROM EMP;

MINIMUM SALARY
--------------
           800
```

26) WRITEAQUERYTOFINDTHEMAXIMUMVALUE OF SALARY OF EMP TABLE.

SQL>SELECTMAX(SAL)“MAXIMUMSALARY”FROMEMP;

```
SQL> SELECT MAX(SAL) “MAXIMUM SALARY” FROM EMP;

MAXIMUM SALARY
--------------
          5000
```

27) WRITEAQUERYTOIMPLEMENTCOUNT FUNCTION ON EMP TABLE.

SQL>SELECTCOUNT(*)“TOTALNOOFROWS”FROMEMP;

```
SQL> SELECT COUNT(*) “TOTAL NO OF ROWS” FROM EMP;

TOTAL NO OF ROWS
----------------
              14
```

28) WRITEAQUERYTOIMPLEMENTSUM FUNCTION ON EMP TABLE.

SQL>SELECTSUM(SAL)”SUMOFSLARY”FROMEMP;

```
SQL> SELECT SUM(SAL)”SUM OF SLARY” FROM EMP;

SUM OF SLARY
------------
       29025
```

PL/SQLQUERIES

1. ADD2NUMBERS.

```
SQL>SETSERVEROUTPUTON;
DECLARE
VAR1INTEGER;
VAR2INTEGER;
VAR3INTEGER;
BEGIN
VAR1    :=
&VAR1;VAR2    :=
&VAR2;VAR3:=VAR1+
VAR2;
DBMS_OUTPUT.PUT_LINE('THESUM     OF'||VAR1    ||'AND    '||VAR2    ||    'IS:'||VAR3);
END;
/
```

OUTPUT:

```
Enter value for var1: 10
old   6:   VAR1 := &VAR1;
new   6:   VAR1 := 10;
Enter value for var2: 20
old   7:   VAR2 := &VAR2;
new   7:   VAR2 := 20;
The sum of 10 and 20 is: 30

PL/SQL procedure successfully completed.
```

2. FORLOOPDEMO

```
SQL>SETSERVEROUTPUT
ON;    DECALARE
    INUMBER(2);
    BEGIN
    FORIIN1..10
    LOOP
    DBMS_OUTPUT.PUT_LINE(
    I);    END    LOOP;
    END;
```

OUTPUT:

```
1
2
3
4
5
6
7
8
9
10

PL/SQL procedure successfully completed.
```

3. PL/SQLPROGRAMTOSWAPTWONUMBERS

i. USINGTEMP

```
SQL>SETSERVEROUTPUTON;
DECLARE
 ANUMBER(2):=&A;
BNUMBER(2):=&B;
TEMP
NUMBER(2);
BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE    SWAPPING
 A='||A||'B='||B);    TEMP:=A;
B:=TEMP;
DBMS_OUTPUT.PUT_LINE('AFTER    SWAPPING
 A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 10
old   2:   A NUMBER(2):=&A;
new   2:   A NUMBER(2):=10;
Enter value for b: 20
old   3:   B NUMBER(2):=&B;
new   3:   B NUMBER(2):=20;
BEFORE SWAPPING A=10B=20
AFTER SWAPPING A=10B=10

PL/SQL procedure successfully completed.
```

ii. WITHOUTUSINGTEMP

```
SQL>SETSERVEROUTPUTON;

 DECLARE
1.  NUMBER(2):=&A;
2.  NUMBER
 (2):=&B;    BEGIN
DBMS_OUTPUT.PUT_LINE('BEFORE
SWAPPING’A='||A||'B='||B);
A:=A+B;    B:=A-B;
A:=A-B;
DBMS_OUTPUT.PUT_LINE('AFTER
SWAPPING’A='||A||'B='||B);    END;
/
```

OUTPUT:

```
Enter value for a: 4
old   2:   A NUMBER(2) := &A;
new   2:   A NUMBER(2) := 4;
Enter value for b: 5
old   3:   B NUMBER(2) := &B;
new   3:   B NUMBER(2) := 5;
BEFORE SWAPPING A=4 B=5
AFTER SWAPPING A=5 B=4

PL/SQL procedure successfully completed.
```

4. WRITEAPROGRAMTOPRINTEMPNO,ENAME,JOB,SALARYOFAN EMPLOYEE OF GIVEN EMPNO.

```
SQL>SETSERVEROUTPUTON;
DECLARE
TYPE EMPREC IS RECORD (ENO
EMP.EMPNO%TYPE,NEMP.ENAME%TYPE,J
EMP.JOB%TYPE,S EMP.SAL%TYPE);
R EMPREC;
ENOEMP.EMPNO%TYPE:=&ENO;
BEGIN
SELECTEMPNO,ENAME,JOB,SAL
INTO R
FROMEMP
WHERE EMPNO = ENO; -- Corrected the
condition
DBMS_OUTPUT.PUT_LINE(R.ENO||''||R.N||''||R.J
||''||R.S);
END;
/
```

OUTPUT:

```
Enter value for eno: 7839
old   4: ENO EMP.EMPNO%TYPE := &ENO;
new   4: ENO EMP.EMPNO%TYPE := 7839;
7839 KING PRESIDENT 5000

PL/SQL procedure successfully completed.
```

5. PL/SQLPROGRAMTOIMPLEMENT%FOUND

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSOREMP_CUR
ISSELECTENAME,COMM
FROMEMPWHEREEMPNO=7654
AND    COMM    IS    NOT    NULL;
EMP_NAME    VARCHAR2(20);
EMP_COMM    NUMBER(10,2);
BEGIN    OPEN    EMP_CUR;
FETCHEMP_CURINTOEMP_NAME,EMP_COMM;
IF    EMP_CUR%FOUND    THEN
DBMS_OUTPUT.PUT_LINE('EMPLOYEE:'||EMP_NAME     ||'GETSCOMMISSION:'||
EMP_COMM);
ELSE
DBMS_OUTPUT.PUT_LINE('EMPLOYEEDIDNOTGETANYCOMMISSION');
END    IF;
DBMS_OUTPUT.PUT_LINE(EMP_CUR%ROWCOUNT||'RECORDSFETCHEDSOFAR');     CLOSE
EMP_CUR;
END;
/
```

OUTPUT

```
EMPLOYEE: MARTIN GETS COMMISSION: 1400
1 RECORDS FETCHED SO FAR

PL/SQL procedure successfully completed.
```

6. PL/SQLTOPMPLEMENT%FOUNDUSINGDEPTTABLE.

```
SQL>SETSERVEROUTPUTON;
DECLARE
CURSORDEPTCURISSELECTDEPTNO,DNAMEFROMDEPTWHERE
DEPTNO = 10;
TDEPTCURDEPTCUR%ROWTYPE;
BEGIN
OPENDEPTCUR;
FETCH DEPTCUR INTO
TDEPTCUR; IF
DEPTCUR%FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('DEPTNO:'||TDEPTCUR.DEPTNO||'DNAME:'||
TDEPTCUR.DNAME);
ELSE
DBMS_OUTPUT.PUT_LINE('No department found
withDEPTNO 10'); END IF;
CLOSEDEPTCUR;
END;
/
```

OUTPUT

```
DEPTNO: 10  DNAME: ACCOUNTING

PL/SQL procedure successfully completed.
```

7. PL/SQL BLOCK USINGTHE COURSOR TO DISPLAY DETAILS OF ALL EMPLOYEE'SFROMEMPTABLEWHOSESUMOFSALANDCOMMIS >2000

```
SQL>SETSERVEROUTPUTON;
DECLARE
MEM_PNO    EMP.EMPNO%TYPE;
MEM_NAMEEMP.ENAME%TYPE;
MEM_SAL    EMP.SAL%TYPE;
CURSOR    C1    IS
SELECTEMPNO,ENAME,SAL+NVL(COMM,0)
FROM    EMP
WHERESAL+NVL(COMM,0)>2000;
 BEGIN
OPENC1;
DBMS_OUTPUT.PUT_LINE('EMPNO    ENAME    MSAL');
LOOP
FETCHC1INTOMEM_PNO,MEM_NAME,MEM_SAL;
EXIT    WHEN    C1%NOTFOUND;
DBMS_OUTPUT.PUT_LINE(MEM_PNO||"||MEM_NAME     ||"||MEM_SAL);
END    LOOP;
CLOSEC1;
END;
/
```

<u>OUTPUT</u>

```
EMPNO  ENAME   MSAL
7839  KING  5000
7698  BLAKE  2850
7782  CLARK  2450
7566  JONES  2975
7902  FORD  3000
7654  MARTIN  2650

PL/SQL procedure successfully completed.
```

8. **PL/SQL BLOCK THE RETRIVES COURSE TABLE INTO CURSOR.INPUT CODEIDTOSEARCH.IFCOURSEEXISTPRINTTHEINFORMATION.IF COURSE ID DOES NOT EXIST, PRINT USER DEFINED MESSAGE BY THROWING USER DEFINED EXCEPTION.**

```
SQL>SETSERVEROUTPUTON;
DECLARE
  C_NAMECOURSE.COURSENAME%TYPE;
COURSORC1ISSELECTCOURSENAMEFROMCOURSEWHERE
COURSEIS:='&ID';
COURSE_NOT_FOUND
EXCEPTION;

BEGIN
OPENC1;
FETCHC1INTOC_NAME;
IF  C1%  FOUND  THEN
DBMS_OUTPUT.PUT_LINE('COURSE  NAME
IS'||C_NAME);
ELSERAISECOURSE_NOT_FOUND;
END  IF;
EXCEPTIONWHENCOURSE_NOT_FOUNDTHEN
DBMS_OUTPUT.PUT_LINE('COURSE  ID  NOT
FOUND');
END;
/
```

<u>OUTPUT</u>

```
Enter value for id: 13
old 3: cursorcl is select coursename from course where courseis=&id'
new 3:cursor cl is select coursename from course where courseis='13';
course name isB.Sc.computeres
PL/SQL procedure successfully completed.
SQL>/
Enter value for id: 18
old 3: cursor cl is select coursename from course where courseis=&id';
new 3:cursor c1 is select coursename from course where courseis=18';
course id is not found
PL/SQL procedure successfully completed
```

9. PL/SQLPROGRAMTOHANDLEZERODIVIDEEXCEPTION

```
SQL>SETSERVEROUTPUTON;
DECLARE
xNUMBER:=&X;
Y       NUMBER
:=   &Y;   BEGIN
IF   Y   <>   0   THEN
DBMS_OUTPUT.PUT_LINE('DIVISION='||X/
Y);   ELSE
DBMS_OUTPUT.PUT_LINE('CHECK    THE
DENOMINATOR');   END   IF;
END;
 /
```

OUTPUT:

```
Enter value for x: 23
old   2:   X NUMBER := &X;
new   2:   X NUMBER := 23;
Enter value for y: 9
old   3:   Y NUMBER := &Y;
new   3:   Y NUMBER := 9;
DIVISION=2.5555555555555555555555555555555555556

PL/SQL procedure successfully completed.
```

10. PL/SQLTHATCREATESATRIGGERTHATINSERTSORUPDATES THE VALUES OF ENAME.JOB AS LOWER-CASE STRING.

```
SQL> CREATE OR REPLACE TRIGGER
LOWERNAMEBEFOREINSERTORUPDATEONEMP
FOR EACH ROW BEGIN
:NEW.ENAME:=:LOWER(“NEW.ENAME);
NEW.JOB:=LOWER(:MEW.JOB);
END;
/
```

OUTPUT:

Triggercreated.

SQL>UPDATEEEMPSETJOB=’MANAGER’WHEREENAME=SMITH;

```
SQL>  UPDATE EMP SET JOB=’MANAGER’ WHERE ENAME='SMITH';

1 row updated.
```

11. PL/SQLFORSIMPLELOOP.

```
SQL>SETSERVEROUTPUTON;
DECLARE
INUMBER:=0;
BEGIN
LOOP
DBMS_OUTPUT.PUT_LINE('I='||I);
I:=I+2;
EXITWHENI>10;
END LOOP;
END;
/
```

OUTPUT:

```
I=0
I=2
I=4
I=6
I=8
I=10

PL/SQL procedure successfully completed.
```